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NOOB TECH DOCUMENTATION

NOOB TECH DOCUMENTATION

Uitenhage Airport

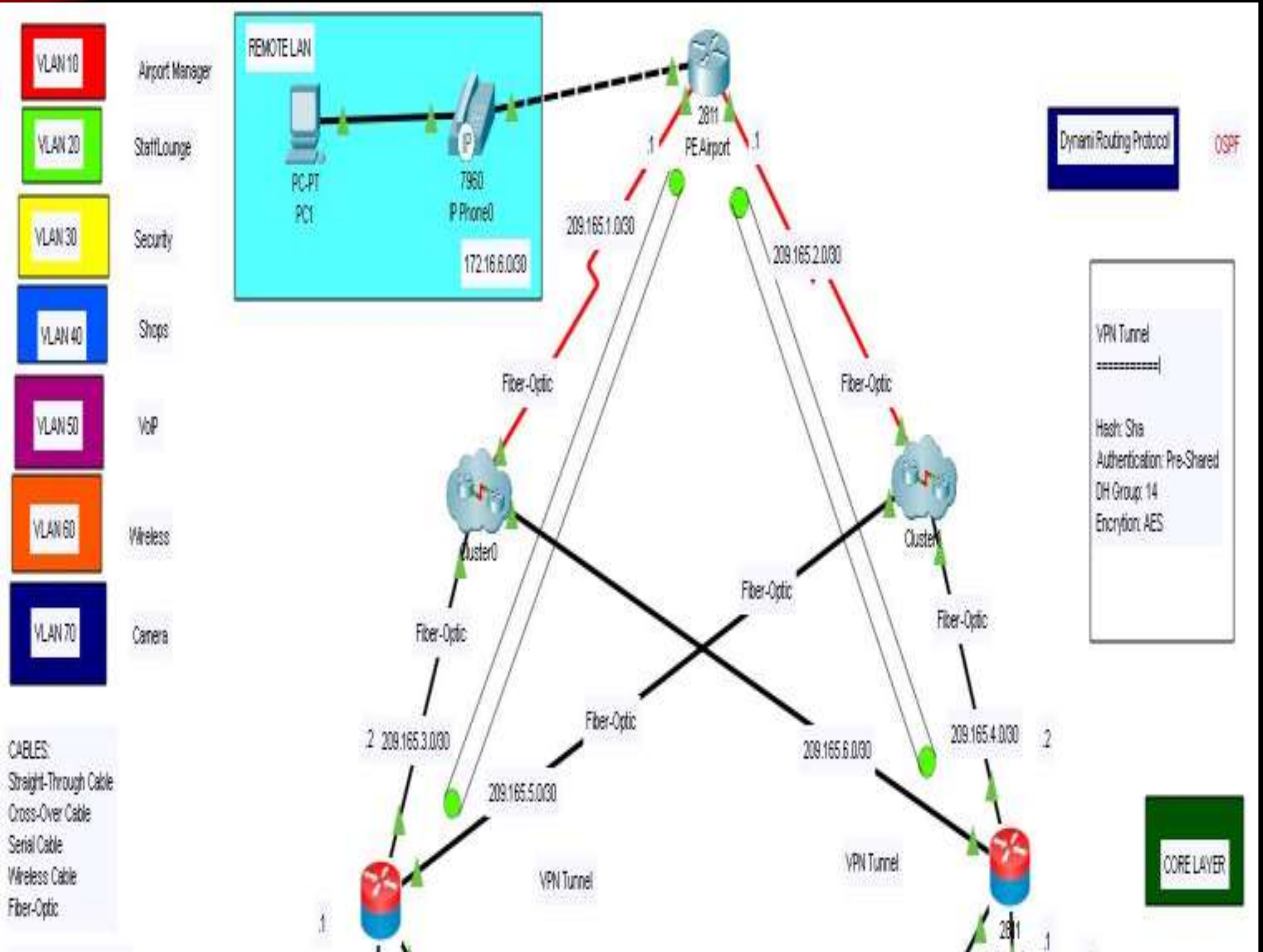
1. Project Scenario
2. Logical Topology
3. Device List
4. Physical Topology
5. Cut Sheet
6. Gantt Chart
7. Configurations

1. PROJECT SCENARIO

- The airport has two ISP's, and is connected to the internet via those ISP's.
- The two border routers configured with OSPF, and they are peering with ISP's. Zone Based is configured for our network protection from outside networks. Connection is secured between remote sites(VPN). Private addresses are translated to public addresses(PAT).
- The access layer switch is connected to the distributed layer switch.
- The headquarter is connected to the internet via ISP
- Passengers need a way to connect to the wireless access.

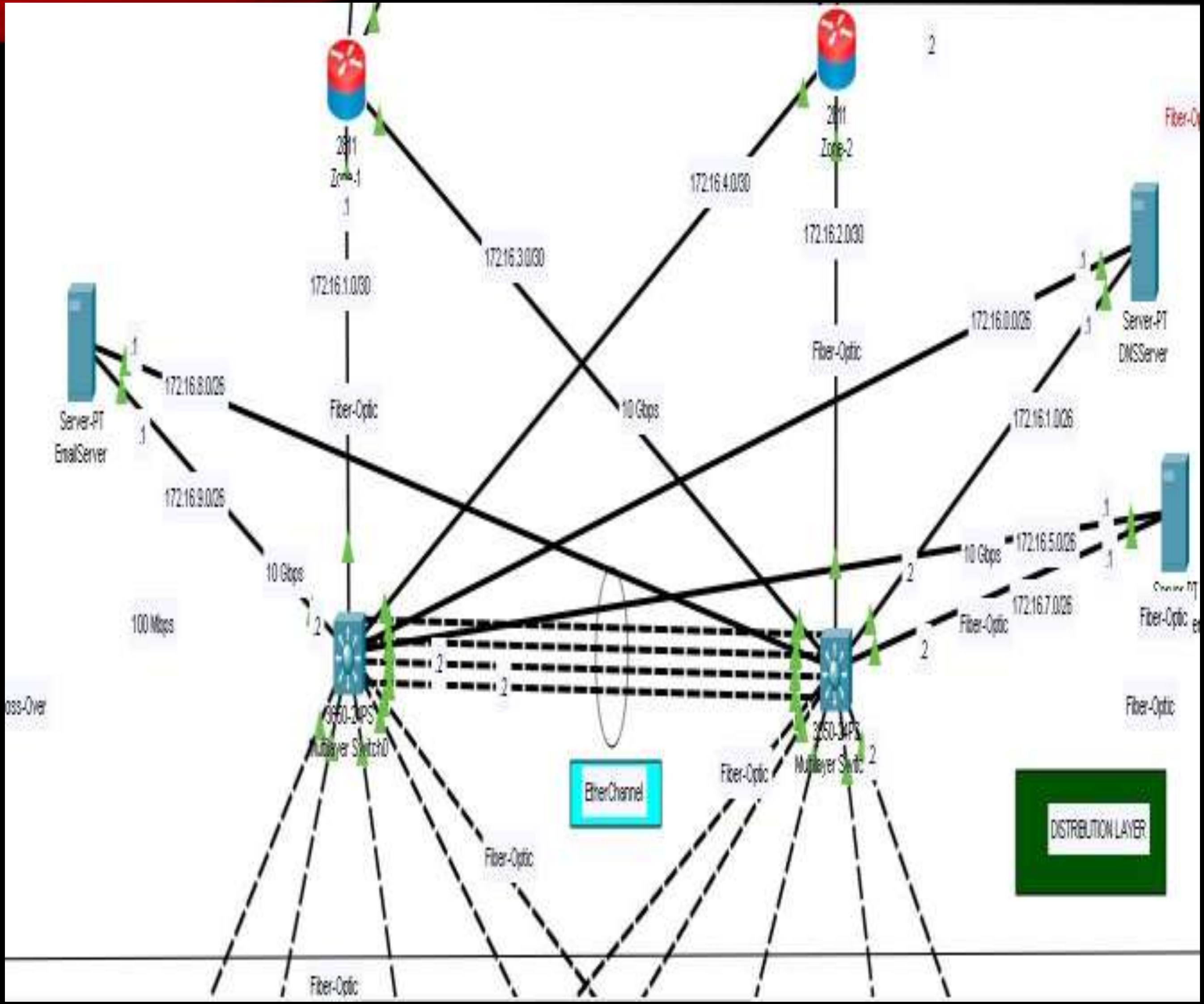
2. LOGICAL TOPOLOGY

CORE LAYER



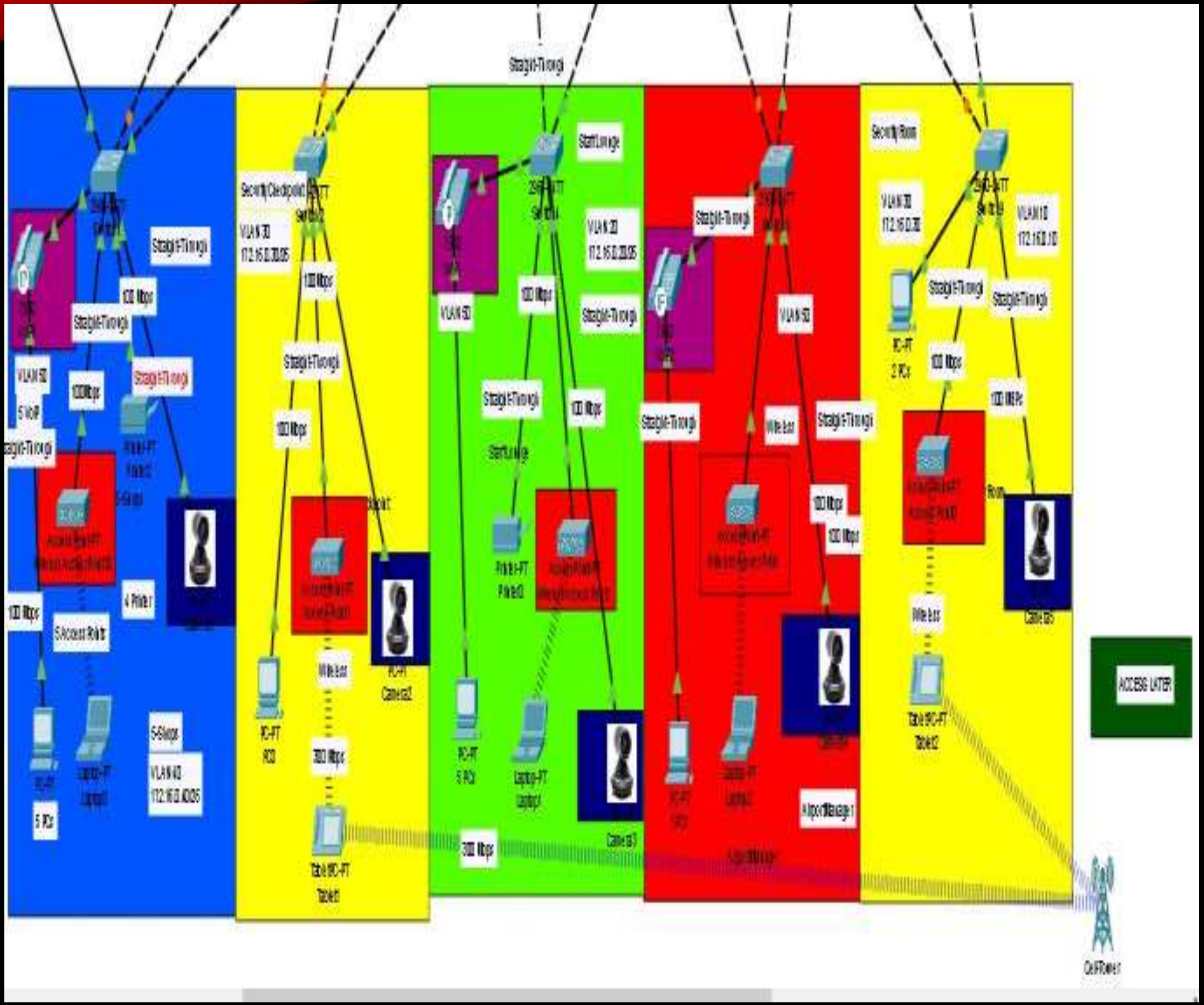
- Two ISP's will provide connectivity to the internet for the airport

Distribution layer



- The two border router will be responsible for routing the traffic between multilayer switches and internet.
- The two multilayer switches will provide connectivity between the access layer switches and the border routers.
- The three servers will be connected to the multilayer switches and they will provide various services like network, example DNS.

ACCESS LAYER



- The access layer switches will provide connectivity for all devices inside VLANs.
- Access points will use access layer switches to provide wireless connectivity to the airport network.

3. DEVICE LIST

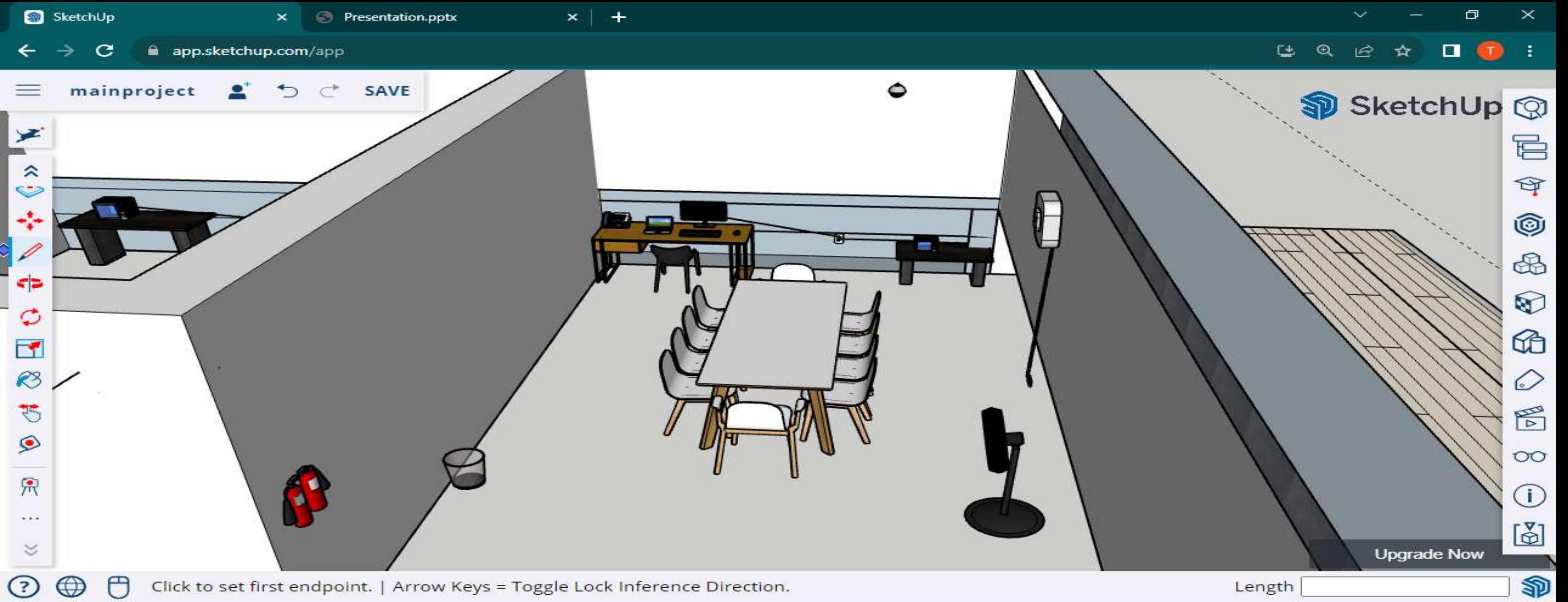
DEVICE LIST FOR LOGICAL TOPOLOGY

NETWORKING EQUIPMENT	MODEL	PRICE	REASON
1. Router	Cisco ISR 2811 Series	R1100	Reliability, security features, and scalability, making them suitable for medium to large enterprises. And also managing traffic between devices.
2. Switches	Cisco Catalyst 2960 Series	R15500	Connects devices in network to each other, by enabling them to talk by, exchanging data.
3. Multilayer Switches	Cisco MDS 3560 Series	R27000	For high-speed. It has dynamic routing protocol
4. Firewalls	Palo Alto Networks PA-220	R24950	Block unauthorized access to the network.
5. Wireless Controller	Cisco Air 2504 Wireless LAN Controller	R14500	Manages wireless network access points and allows wireless devices to connect to the network.
6. Access Points	Ubiquiti Unifi WIFI 6 LR Access Point	R3300	To create a wireless local area network.
7. VPN	Cisco ASA 5506-X	R28000	For secure connection between user and internet.
8. Cameras	Hikvision 4MP AcuSense	R2300	To keep track of the interior and exterior of a property, transmit the signal to a monitor or set of monitors, and give real-time 24/7 viewing access.
9. IP Phones	Cisco CP-7960 IP Phone	R4200	To support voice calling.
10. Server	Dell PowerEdge T150 Server	R26050	Store, receive, and send data.
11. PCs	Dell Optiplex 5400AIO I7	R4500	Store, retrieve and process data of all kinds.
12. Kiosks	CTA Digital Premium Locking Floor Stand Kiosk	R5060	To reach their customers in a more simple and informal manner.
13. Printers	Epson L3251 A4 Multifunction Colour Printer with Wi-Fi Direct	R5400	To present text or images to flat media such as paper in various sizes.
14. Fiber-Optic Cable	10Gb/s QSFP28 Active Optical Cable	R3600	Used for long-distance and high-performance data networking.

15. Straight-Through Cable	RJ45 UTP Cable	R140	Used for connecting unlike devices.
16. Wireless cable	TP-LINK 300Mbps Wi-Fi USB Adapter	R189	Allows you to connect wirelessly to high-speed internet connections.
17. Cross-Over	CAT-5 1 Meter Cross-Over Network Cable	R100	To connect similar devices directly together.

4. PHYSICAL TOPOLOGY

STAFF ROOM


SketchUp
Presentation.pptx
app.sketchup.com/app
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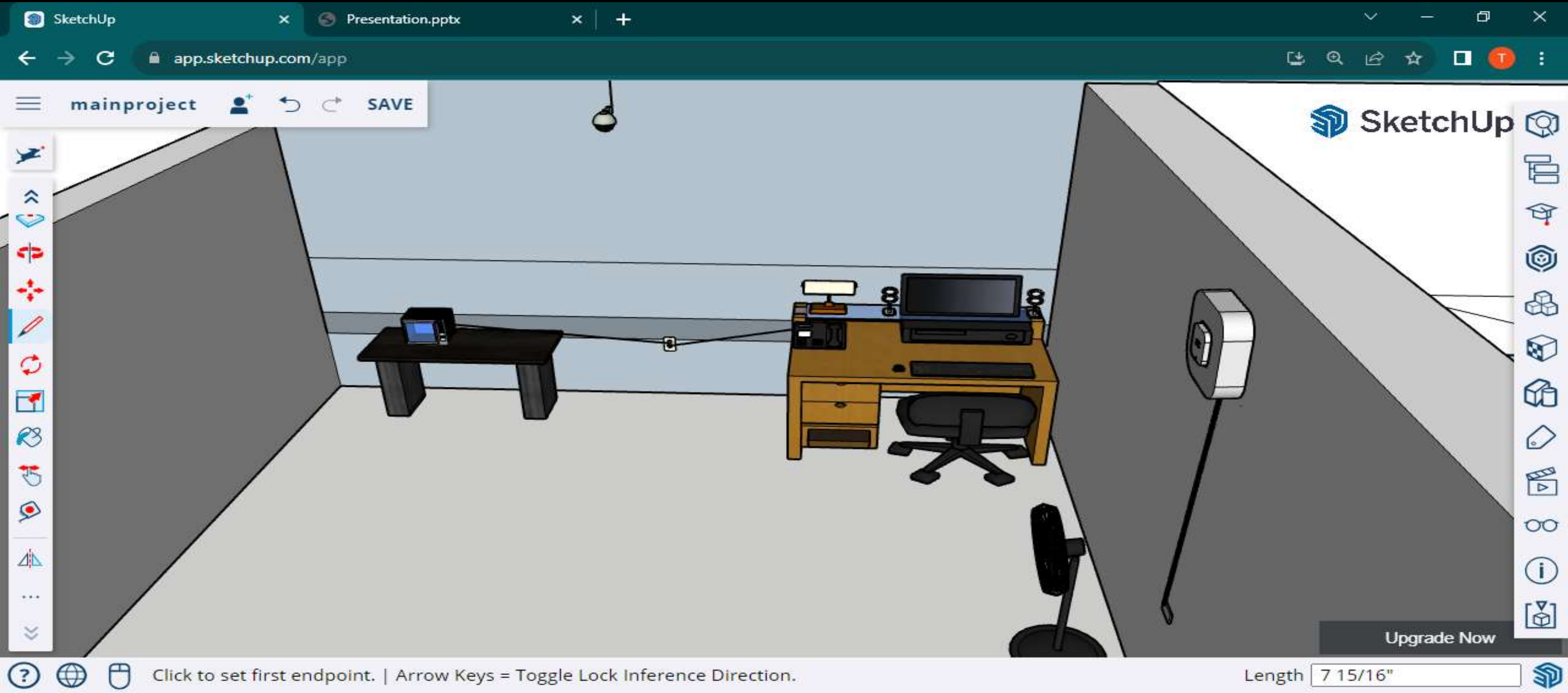
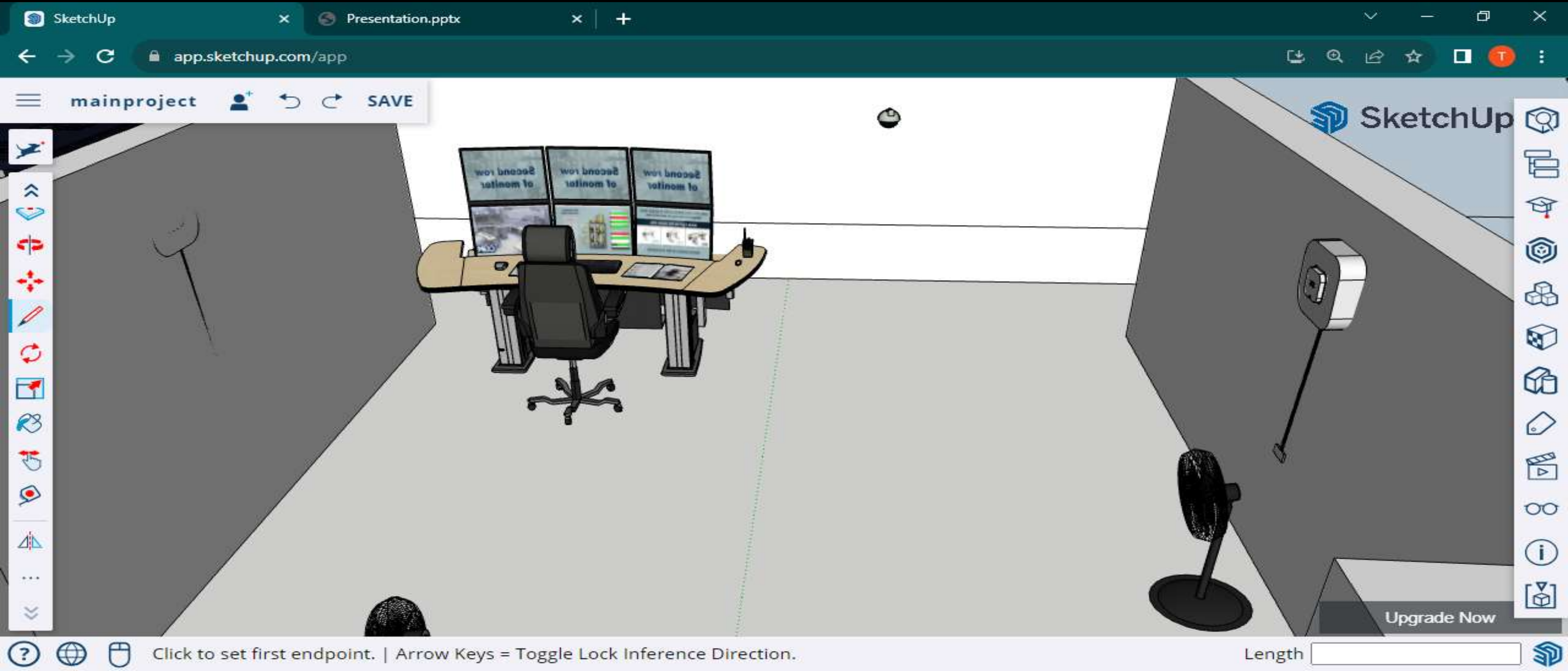
SAVE
SketchUp
Upgrade Now
Click to set first endpoint. | Arrow Keys = Toggle Lock Inference Direction.
Length

SECURITY CHECKPOINT

MANAGER ROOM



SECURITY ROOM



SERVER ROOM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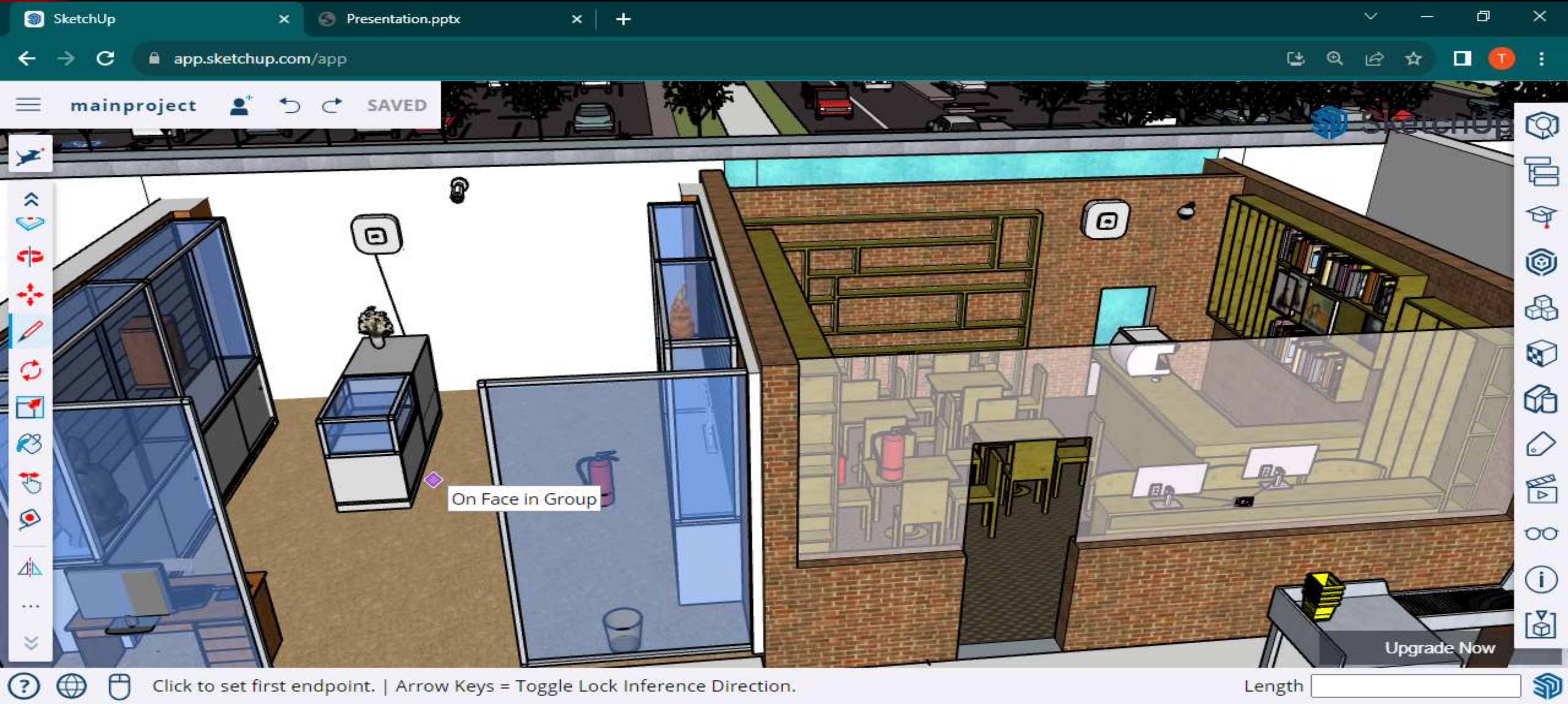
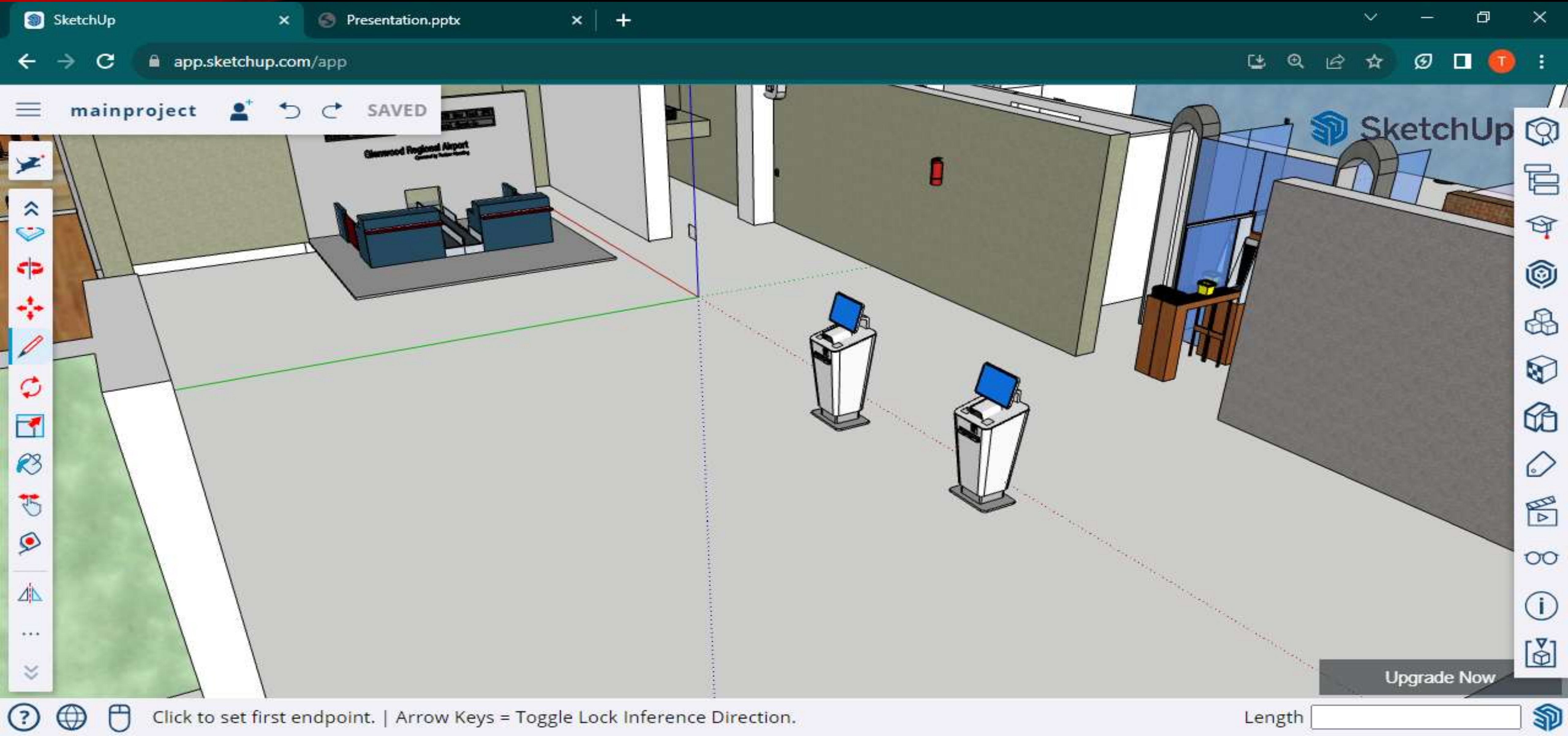
SketchUp
Upgrade Now
Drag in direction to pan
Measurements

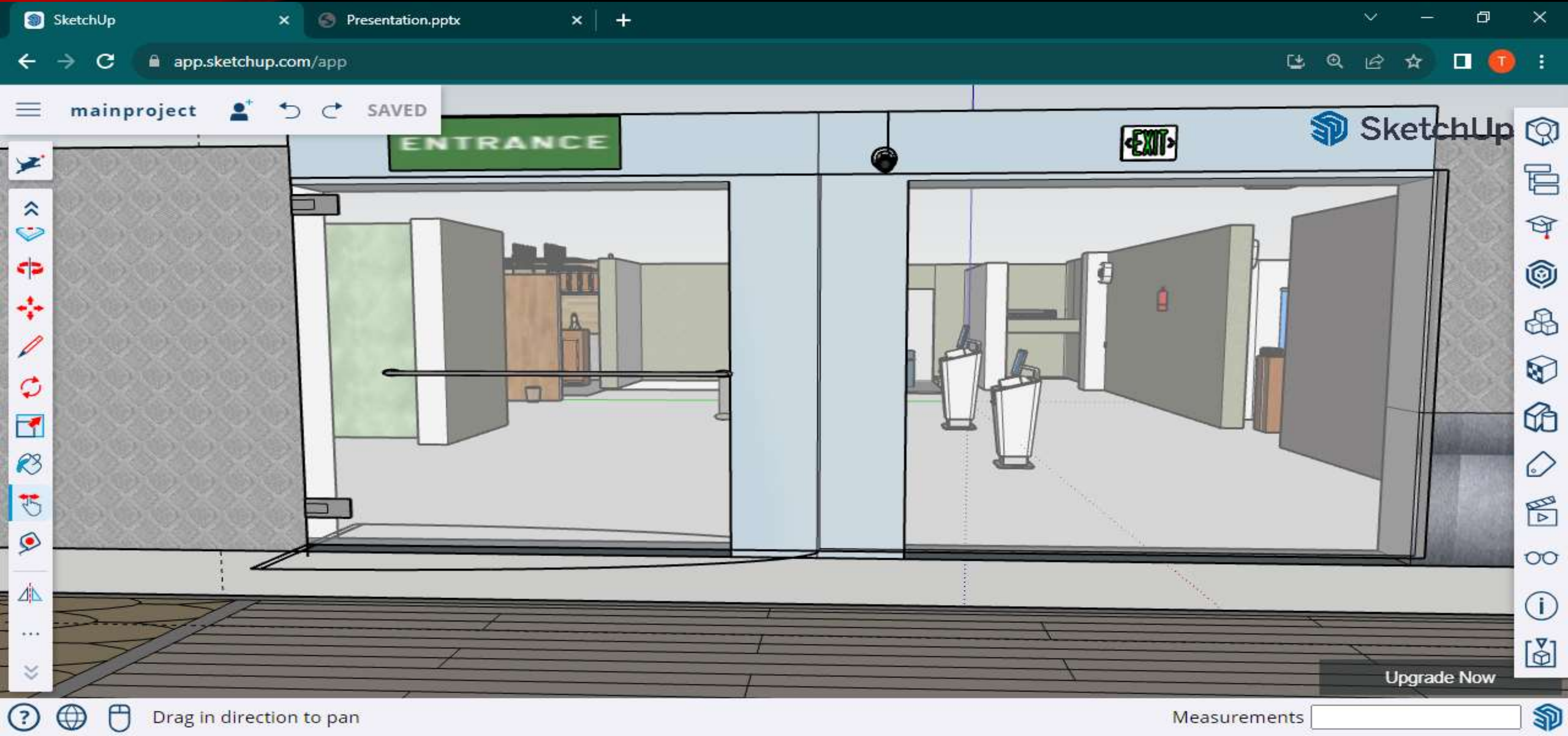
BOOK AND GIFT SHOPS



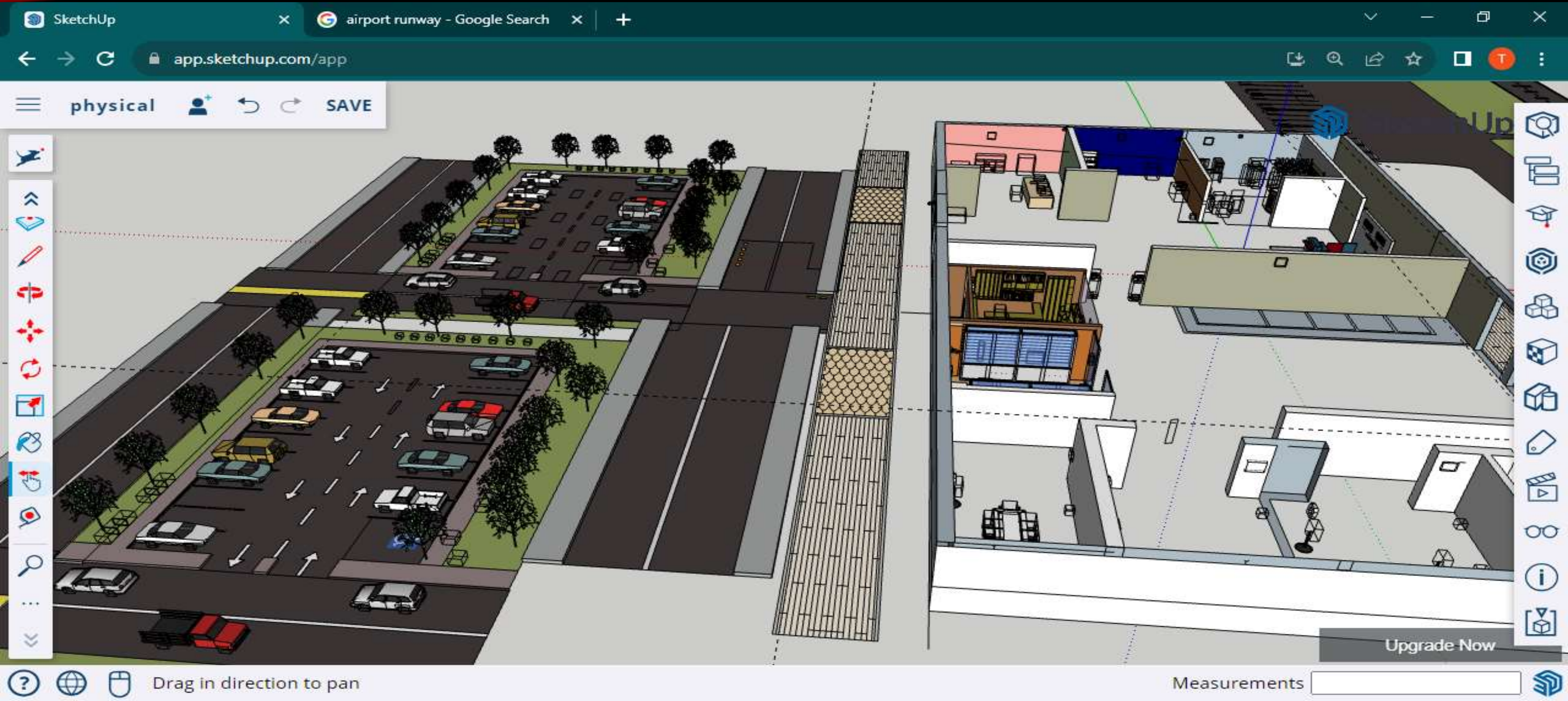
SELF-CHECK-IN



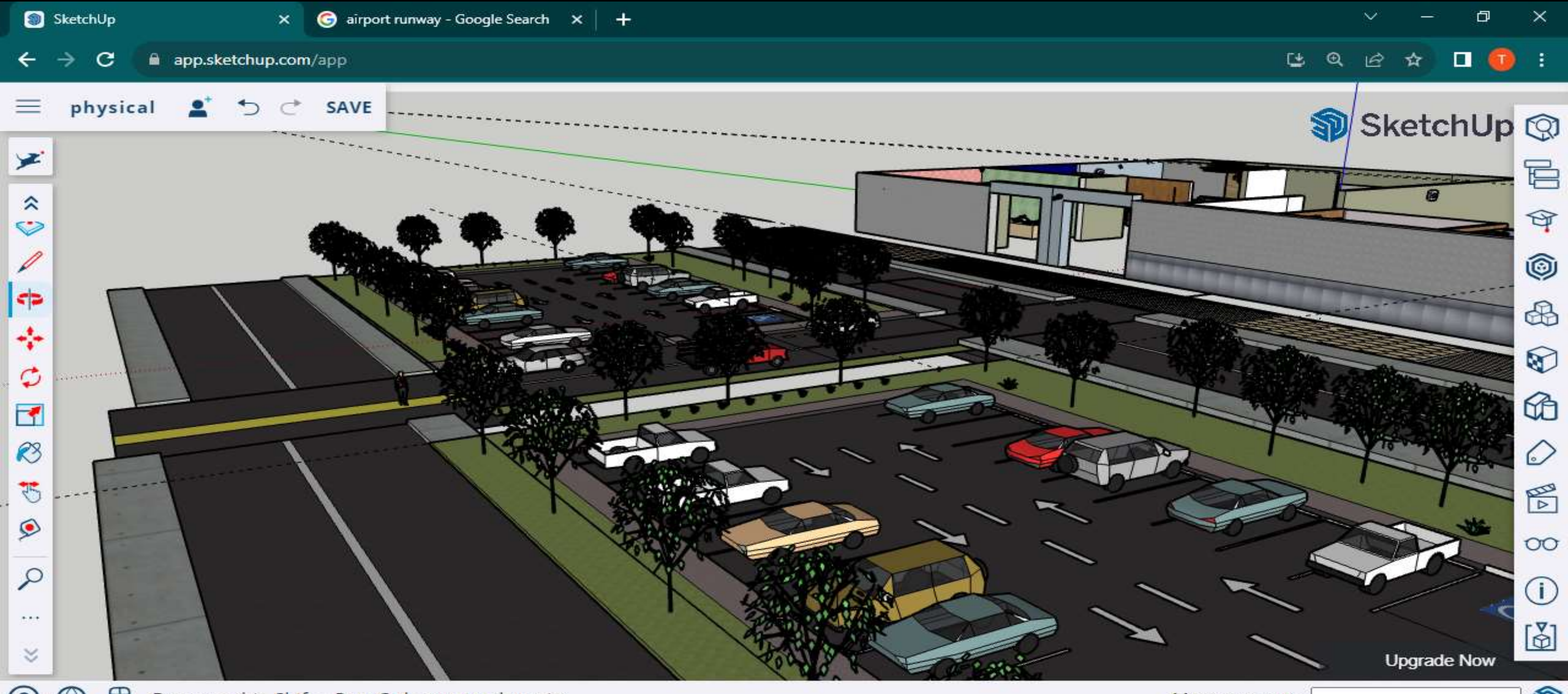
MAIN ENTRANCE



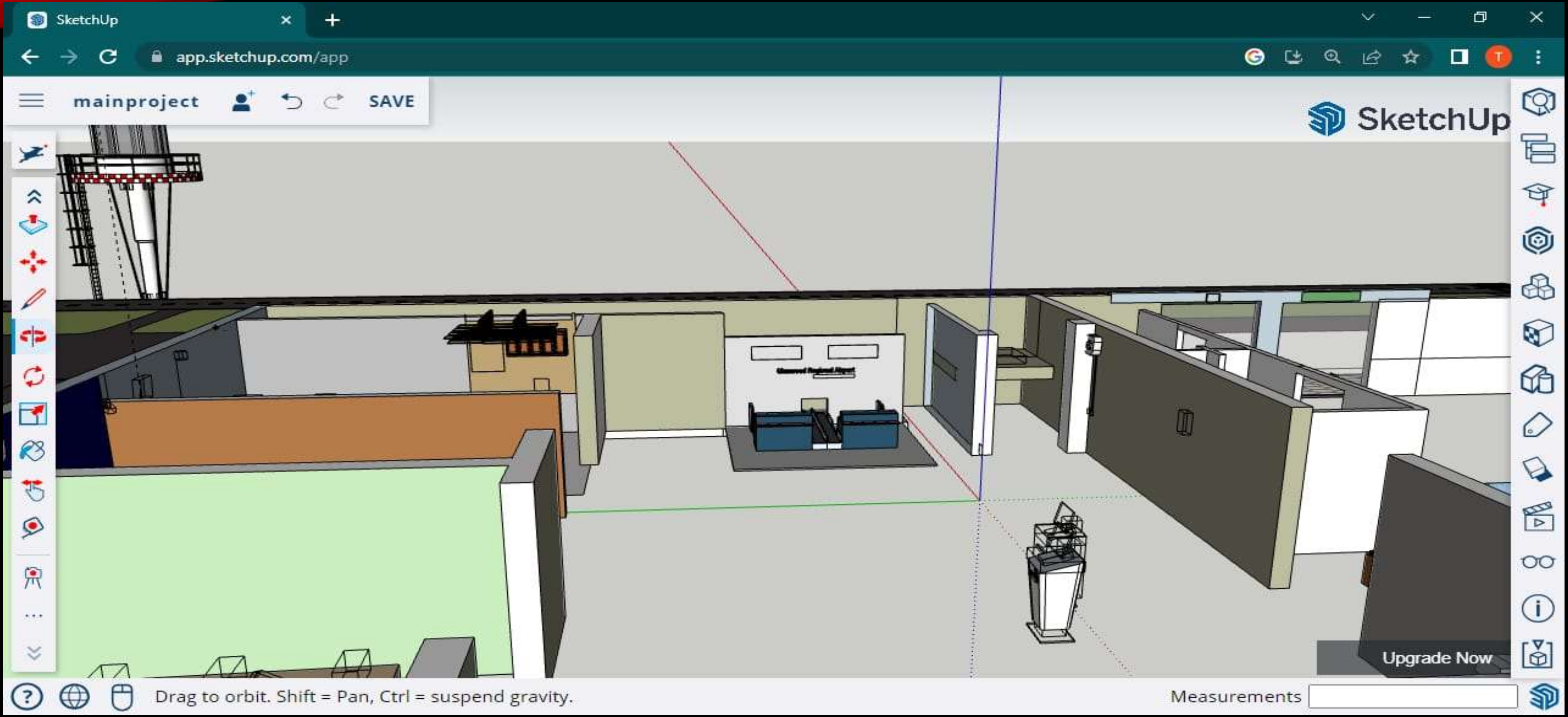
OVERVIEW OF THE TOPOLOGY



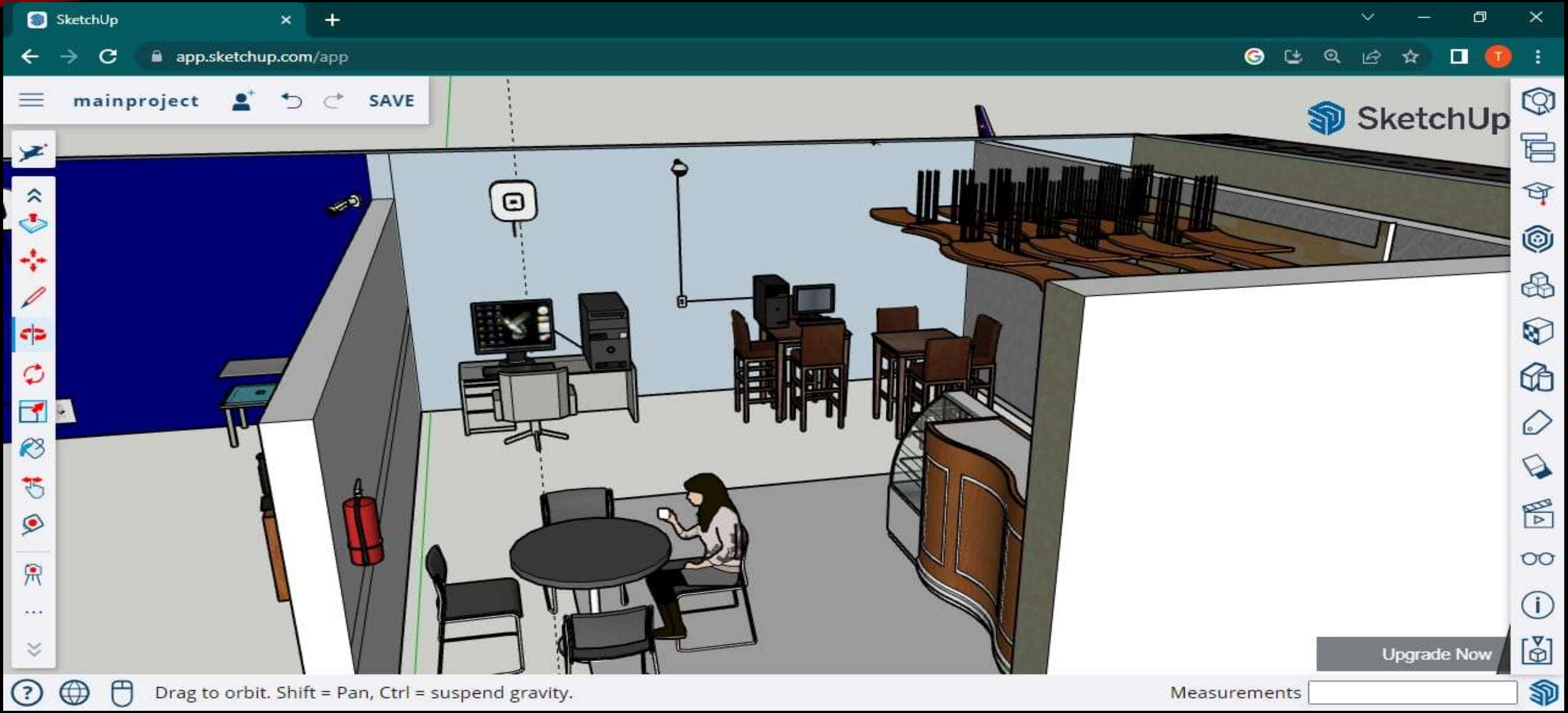
PUBLIC PARKING



PUBLIC AREA


SketchUp
app.sketchup.com/app
mainproject
SAVE
SketchUp
Upgrade Now
Drag to orbit. Shift = Pan, Ctrl = suspend gravity.
Measurements

COFFEE SHOP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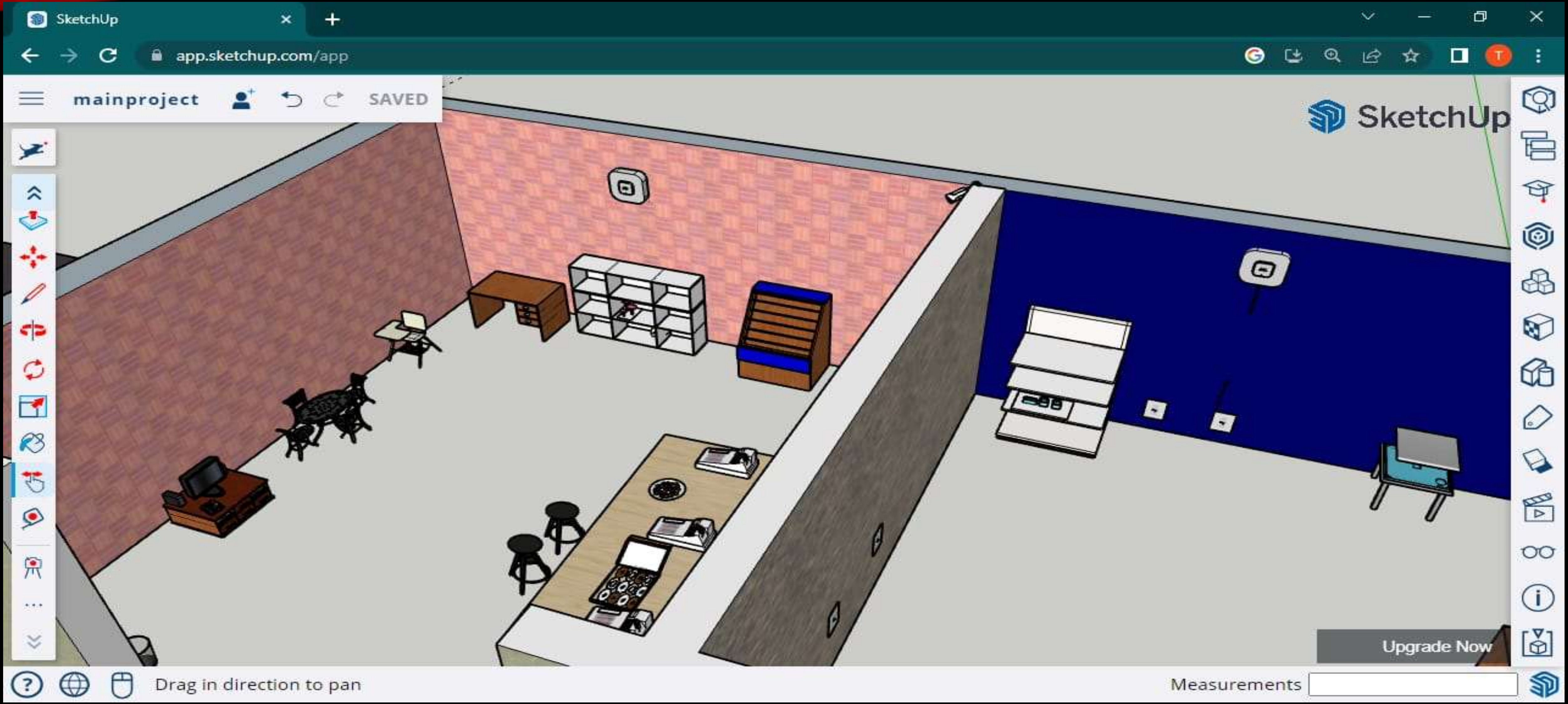
SketchUp
Upgrade Now
Drag to orbit. Shift = Pan, Ctrl = suspend gravity.
Measurements

SWEET SHOP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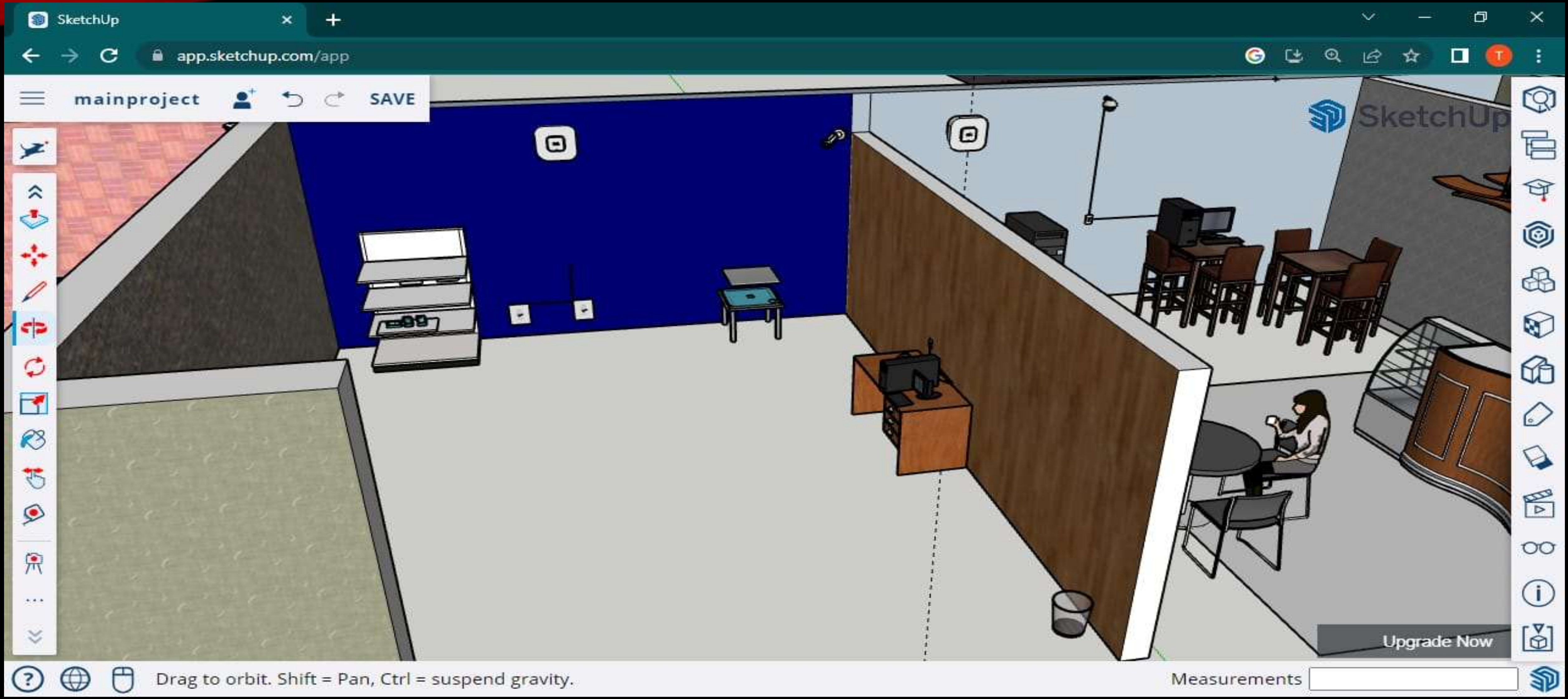
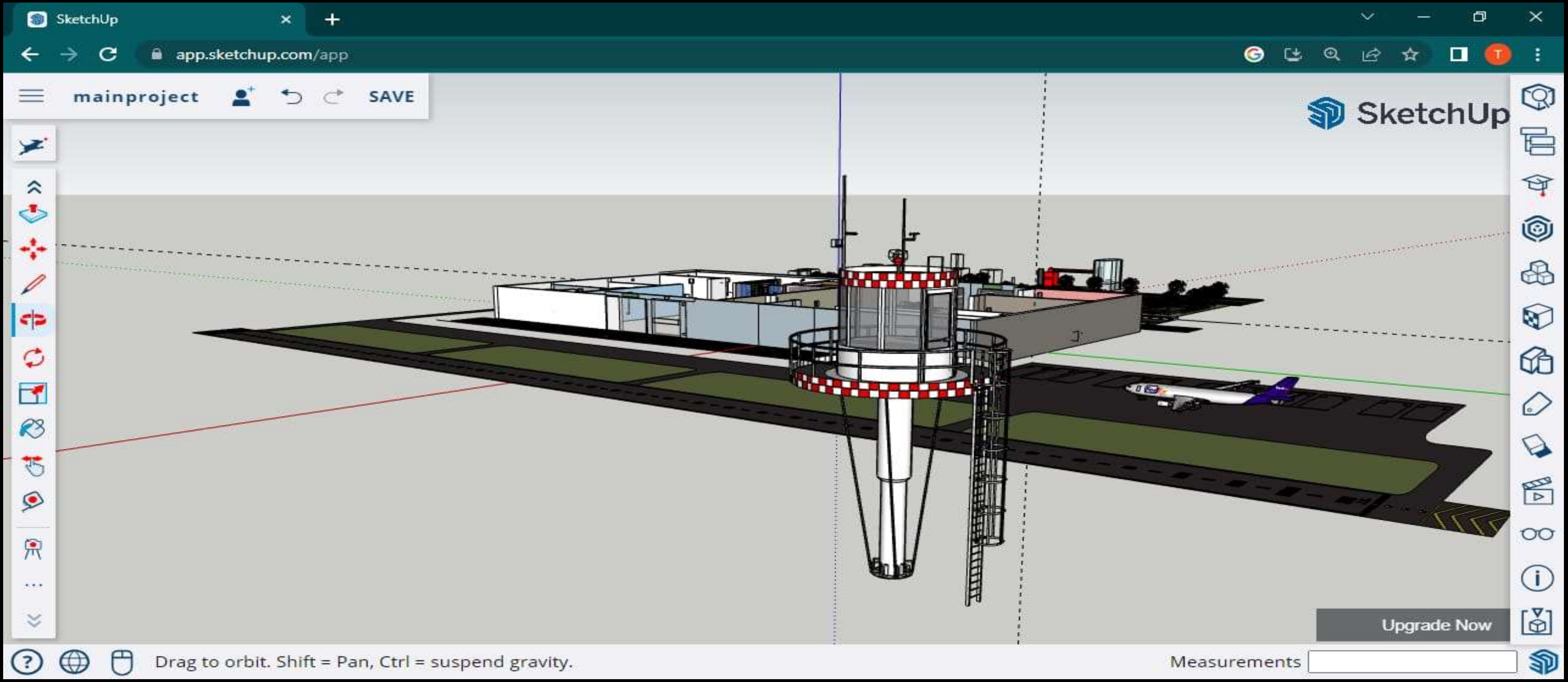
SketchUp
Upgrade Now
Drag in direction to pan
Measurements

CELLPHONE OUTLET



CELL TOWER



SketchUp
app.sketchup.com/app
mainproject
SAVE
SketchUp
Upgrade Now
Drag to orbit. Shift = Pan, Ctrl = suspend gravity.
Measurements

5. CUT SHEET

SERVER ROOM

Switch Port Number1	Switch Port Number2	Dinstance Cable(m)	Cable Name
Server-MLS1g1/0/1	Staff-SW_F0/1	12.2	Staff-SW_F0/1<>Server-MLS1g1/0/1
Server-MLS1g1/0/2	Security-SW_F0/1	11.2	Security-SW_F0/1<>Server-MLS11/0/2
Server-MLS1g1/0/3	Cell-SW_F0/1	17.3	Cell-SW_F0/3<>Server-MLS1G1/0/3
Server-MLS2g1/0/12	Staff-SW_F0/2	12.5	Staff-SW_F0/2<>Server-MLS1G1/0/12
Server-MLS2g1/0/2	Security-SW_F0/2	11.5	Security-SW_F0/2<>Server-MLS2g1/0/2
Server-MLS2G1/0/23	Cell-SW_F0/6	17.8	Cell-SW_F0/2<>Server-MLS1G1/0/23
Server-MLS1G1/0/8	DNS-F0/0	6.2	DNSF0/0<>Server-MLS1G1/0/8
Server-MLS1G1/0/9	DHCP-F0/0	6.6	DHCPF0/0<>Server-MLS1G1/0/9
Server-MLS1G1/0/10	EMAILF0/0	6.9	EMAIL-F0/0<>Server-MLS1G1/0/10
Server-MLS2G1/0/8	DNS-F0/1	6.4	DNSF0/1<>Server-MLS2G1/0/8
Server-MLS2G1/0/9	DHCP-F0/1	6.8	DHCPF0/1<>Server-MLS2G1/0/9
Server-MLS2G1/0/10	EMAILF0/1	7.1	EMAIL-F0/1<>Server-MLS2G1/0/10
Total		**122.5**	

color	cable description	cable name	cable length
grey	switchport trunk	fiber optic	122.5

STAFF ROOM

Switch Port NO.	PC Number	Pc Name	Distance	Cable Name
Staff-SW_F0/1	PC1	Staff-PC1	9.3	Staff-PC1<>Staff-SW_F0/1
Staff-SW_F0/2	PC2	Staff-PC2	8.9	Staff-PC2<>Staff-SW_F0/2
Staff-SW_F0/3	PC3	Staff-PC3	8.3	Staff-PC3<>Staff-SW_F0/3
Staff-SW_F0/4	PC4	Staff-PC4	7.9	Staff-PC4<>Staff-SW_F0/4
Staff-SW_F0/5	PC5	Staff-PC5	5.8	Staff-PC5<>Staff-SW_F0/5
Staff-SW_F0/6	PC6	Staff-PC6	6.2	Staff-PC6<>Staff-SW_F0/6
Staff-SW_F0/7	PC7	Staff-PC7	7.8	Staff-PC7<>Staff-SW_F0/7
Staff-SW_F0/8	PC8	Staff-PC8	8.2	Staff-PC8<>Staff-SW_F0/8
Staff-SW_F0/9	PC9	Staff-PC9	9.1	Staff-PC9<>Staff-SW_F0/9
Staff-SW_F0/10	PC10	Staff-PC10	9.4	Staff-PC10<>Staff-SW_F0/10
Total		**10**	**80.9**	

color	cable description	cable name	cable length
grey	switchport trunk	straight-through cat 5	80.9

MANAGER ROOM

Switch Port Number	PC Number	PC Name	Distance Cable(m)	Cable Name
Security-SW_F0/5	PC1	Manager-PC1	12.2	Manager-PC1<>Security-SW_F0/5
Security-SW_F0/6	IP Phone1	Manager-IP Phone1	12.7	Manager-IP Phone1<>Security-SW_F0/6
Total	2		24.9	

color	cable description	cable name	cable length
grey	switchport trunk	straight-through cat 5	24.9

SECURITY ROOM

Switch Port Number	PC Number	PC Name	Distance Cable(M)	Cable Name
Security-SW_F0/1	PC1	Security-PC1	6.7	Security-PC1<>Security-SW_F0/1
Security-SW_F0/2	PC2	Security-PC2	6.4	Security-PC2<>Security-SW_F0/2
Security-SW_F0/3	PC3	Security-PC3	5.3	Security-PC3<>Security-SW_F0/3
Security-SW_F0/4	PC4	Security-PC4	6.2	Security-PC4<>Security-SW_F0/4
Security-SW_F0/5	IP Phone1	Security-IP Phone1	6.3	Security-IP Phone1<>Security-SW_F0/5
Total	5		30.9	

color	cable description	cable name	cable length
grey	switchport trunk	straight-through cat 5	30.9

GIFTSHOP

3					
4	Switch Port Number	PC Number	PC Name	Distance Cable(m)	Cable Name
5	Staff-SW_F0/11	PC1	Gift-PC1	14.2	Gift-PC1<>Staff-SW_F0/11
6	Staff-SW_F0/12	PC2	Gift-PC2	14.4	Gift-PC2<>Staff-SW_F0/12
7	Staff-SW_F0/13	IP Phone1	Gift-IP Phone1	14.8	Gift-IP Phone1<>Staff-SW_F0/13
8					
9					
10	Total	3		43.4	
11					
12					
13	color	cable description	cable name	cable length	
14	grey	switchport trunk	straight-through cat 5	43.4	
15					
16					

SWEET SHOP

Switch Port Number	PC Number	PC Name	Distance Cable(m)	Cable
Cell-SW_F0/7	PC1	Sweet-PC1	12.2	Sweet-PC1<>Cell-SW_F0/7
Cell-SW_F0/8	PC2	Sweet-PC2	12.5	Sweet-PC2<>Cell-SW_F0/8
Cell-SW_F0/9	IP Phone1	Sweet-IP Phone1	12.7	Sweet-IP Phone1<>Cell-SW_F0/9
Total	3		37.4	

color	cable description	cable name	cable length
grey	switchport trunk	straight-through cat 5	37.4

COFFEE SHOP

Switch Port Number	PC Number	PC Name	Distance Cable(m)	Cable Name
Cell-SW_F0/4	PC1	Coffee-PC1	12.1	Coffee-PC1<>Cell-SW_F0/4
Cell-SW_F0/5	PC2	Coffee-PC2	12.5	Coffee-PC2<>Cell-SW_F0/5
Cell-SW_F0/6	IP Phone1	Coffee-IP Phone1	12.9	Coffee-IP Phone1<>Cell-SW_F0/6
Total	3		37.5	

color	cable description	cable name	cable length
grey	switchport trunk	straight-through cat 5	37.5

BOOK SHOP

Switch Port Number	PC Number	PC Name	Distance Cable(m)	Cable Name
Staff-SW_F0/14	PC1	Book-PC1	16.6	Book-PC1<>Staff-SW_F0/14
Staff-SW_F0/15	PC2	Book-PC2	16.4	Book-PC2<>Staff-SW_F0/15
Staff-SW_F0/16	IP Phone1	Book-IP Phone1	16.3	Book-IP Phone1<>Staff-SW_F0/16
Total	3		49.3	

color	cable description	cable name	cable length
grey	switchport trunk	straight-through cat 5	49.3

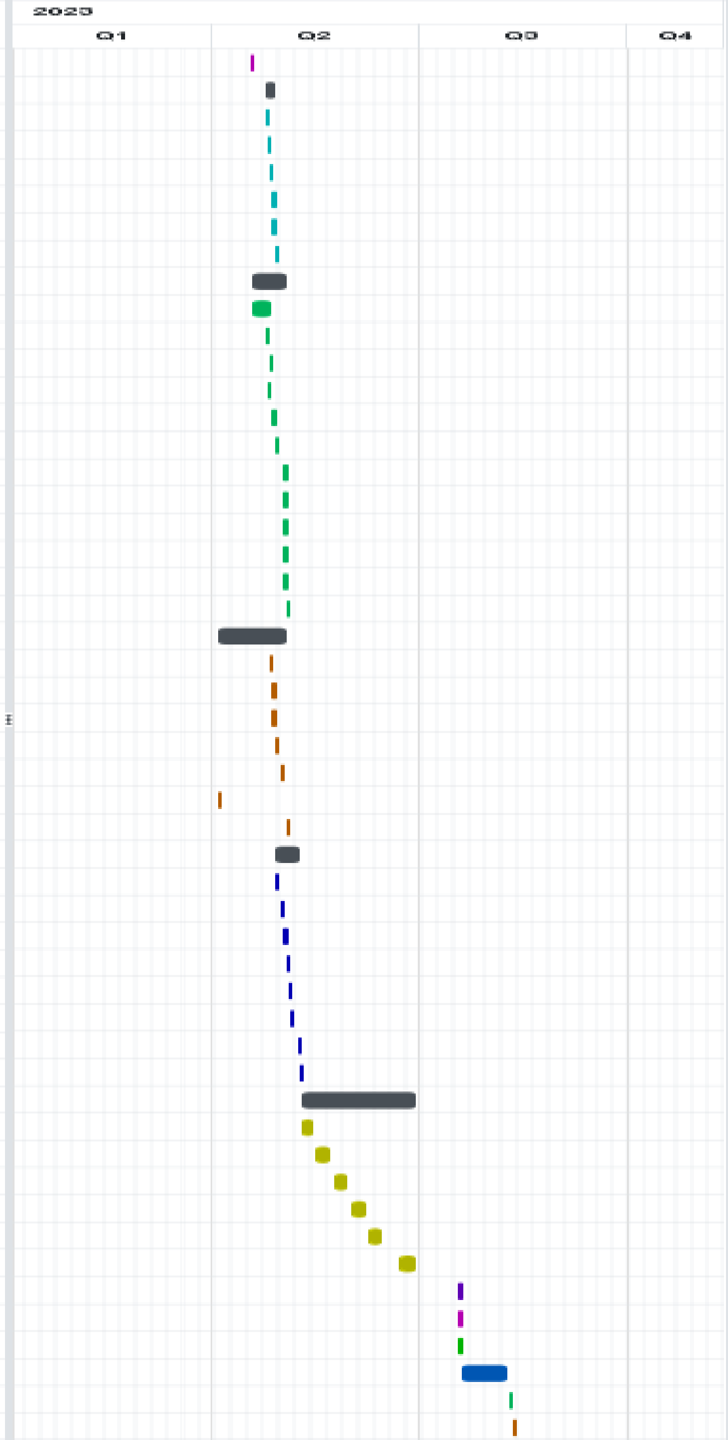
CELLPHONE OUTLET

Switch Port Number	PC Number	PC Name	Distance Cable(m)	Cable Name
Cell-SW_F0/1	PC1	Cell-PC1	7.8	Cell-PC1<>Cell-SW_F0/1
Cell-SW_F0/2	PC2	Cell-PC2	8.1	Cell-PC2<>Cell-SW_F0/2
Cell-SW_F0/3	IP Phone1	Cell-IP Phone1	8.5	Cell-IP Phone1<>Cell-SW_F0/3
Total	3		24.4	

color	cable description	cable name	cable length
grey	switchport trunk	straight-through cat 5	24.4

6. GANTT CHART

Name	Start Date	End Date	Duration	Progress %	Color
Onsite Inspection	Apr 17, 2023	Apr 17, 2023	1 day	100	
Manager Room	Apr 24, 2023	Apr 28, 2023	5 days	94	
Drilling	Apr 24, 2023	Apr 25, 2023	2 days	100	
Cleaning	Apr 25, 2023	Apr 25, 2023	1 day	100	
Trunking	Apr 26, 2023	Apr 26, 2023	1 day	100	
Installing Cables	Apr 27, 2023	Apr 27, 2023	1 day	90	
Installing Access Point	Apr 27, 2023	Apr 27, 2023	1 day	87	
Installing CCTV	Apr 28, 2023	Apr 28, 2023	1 day	85	
Server Room	Apr 18, 2023	May 03, 2023	12 days	97	
Raising Floor	Apr 18, 2023	Apr 26, 2023	7 days	100	
Drilling	Apr 24, 2023	Apr 25, 2023	2 days	100	
Cleaning	Apr 26, 2023	Apr 26, 2023	1 day	100	
Trunking	Apr 25, 2023	Apr 25, 2023	1 day	100	
Installing Cables	Apr 27, 2023	Apr 27, 2023	1 day	100	
Installing Racks	Apr 28, 2023	Apr 28, 2023	1 day	100	
Installing Servers	May 02, 2023	May 02, 2023	1 day	90	
Installing Switches	May 02, 2023	May 02, 2023	1 day	90	
Installing Firewalls	May 02, 2023	May 02, 2023	1 day	90	
Installing Routers	May 02, 2023	May 02, 2023	1 day	90	
Installing Patch Panels	May 02, 2023	May 02, 2023	1 day	90	
Installing CCTV	May 03, 2023	May 03, 2023	1 day	100	
Security Room	Apr 03, 2023	May 03, 2023	23 days	98	
Drilling	Apr 26, 2023	Apr 26, 2023	1 day	100	
Cleaning	Apr 27, 2023	Apr 27, 2023	1 day	100	
Trunking	Apr 27, 2023	Apr 27, 2023	1 day	100	
Installing Cables	Apr 28, 2023	Apr 28, 2023	1 day	100	
Installing Access Point	May 01, 2023	May 01, 2023	1 day	100	
Installing Monitors	Apr 03, 2023	Apr 03, 2023	1 day	100	
Installing CCTV	May 03, 2023	May 03, 2023	1 day	90	
Staff Lounge	Apr 28, 2023	May 09, 2023	8 days	98	
Drilling	Apr 28, 2023	Apr 28, 2023	1 day	100	
Cleaning	May 01, 2023	May 01, 2023	1 day	100	
Trunking	May 02, 2023	May 02, 2023	1 day	100	
Installing Cabes	May 03, 2023	May 03, 2023	1 day	100	
Installing Patch Panels	May 04, 2023	May 04, 2023	1 day	100	
Installing Switch	May 05, 2023	May 05, 2023	1 day	100	
Installing Access Point	May 08, 2023	May 08, 2023	1 day	100	
Installing CCTV	May 09, 2023	May 09, 2023	1 day	89	
Shops	May 10, 2023	Jun 29, 2023	37 days	98	
Drilling	May 10, 2023	May 15, 2023	4 days	100	
Cleaning	May 16, 2023	May 22, 2023	5 days	100	
Trunking	May 24, 2023	May 30, 2023	5 days	100	
Installing Cables	Jun 01, 2023	Jun 07, 2023	5 days	100	
Installing Access Point	Jun 08, 2023	Jun 14, 2023	5 days	100	
Installing CCTV	Jun 22, 2023	Jun 29, 2023	6 days	90	
Installing Kiosks	Jul 18, 2023	Jul 20, 2023	3 days	100	
Installing Air Conditioners	Jul 18, 2023	Jul 20, 2023	3 days	100	
Installing Fire Extinguishers	Jul 18, 2023	Jul 20, 2023	3 days	100	
Configuration and Testing	Jul 19, 2023	Aug 08, 2023	15 days	100	
Cleaning	Aug 09, 2023	Aug 10, 2023	2 days	100	
Documentation	Aug 11, 2023	Aug 11, 2023	1 day	100	



7. CONFIGURATIONS

DHCP

DHCP will provide ip addresses in the devices in vlans


```
!
hostname ML-S1
!
!
!
!
ip dhcp pool VLAN10
 network 172.16.10.0 255.255.255.0
 default-router 172.16.10.3
 dns-server 172.16.0.1
ip dhcp pool VLAN20
 network 172.16.20.0 255.255.255.0
 default-router 172.16.20.3
 dns-server 172.16.0.1
ip dhcp pool VLAN30
 network 172.16.30.0 255.255.255.0
 default-router 172.16.30.3
 dns-server 172.16.0.1
ip dhcp pool VLAN40
 network 172.16.40.0 255.255.255.0
 default-router 172.16.40.3
 dns-server 172.16.0.1
ip dhcp pool VLAN50
 network 172.16.50.0 255.255.255.0
 default-router 172.16.50.3
 dns-server 172.16.0.1
ip dhcp pool VLAN60
 network 172.16.60.0 255.255.255.0
 default-router 172.16.60.3
 dns-server 172.16.0.1
 domain-name wr
!
```

```
!
hostname ML-S2
!
!
!
!
ip dhcp pool VLAN10
 network 172.16.10.0 255.255.255.0
 default-router 172.16.10.2
 dns-server 172.16.0.1
ip dhcp pool VLAN20
 network 172.16.20.0 255.255.255.0
 default-router 172.16.20.2
 dns-server 172.16.0.1
ip dhcp pool VLAN30
 network 172.16.30.0 255.255.255.0
 default-router 172.16.30.2
 dns-server 172.16.0.1
ip dhcp pool VLAN40
 network 172.16.40.0 255.255.255.0
 default-router 172.16.40.2
 dns-server 172.16.0.1
ip dhcp pool VLAN50
 network 172.16.50.0 255.255.255.0
 default-router 172.16.50.2
 dns-server 172.16.0.1
ip dhcp pool VLAN60
 network 172.16.60.0 255.255.255.0
 default-router 172.16.60.2
 dns-server 172.16.0.1
 domain-name wr
!
```

VLANS

Configure vlans in MLS1 and MLS2 to group devices

```
ML-S1#sh vla br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/11, Gig1/0/12, Gig1/0/13,
Gig1/0/14
                                                Gig1/0/15, Gig1/0/16, Gig1/0/17,
Gig1/0/18
                                                Gig1/0/19, Gig1/0/20, Gig1/1/1, Gig1/1/2
                                                Gig1/1/3, Gig1/1/4
10   Manager_Vlan                     active
20   Staff_Vlan                       active
30   Security_Vlan                    active
40   Shop_Vlan                        active
50   VoIP                             active
60   Wireless                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
ML-S1#
```

```
ML-S2#sh vla
ML-S2#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/11, Gig1/0/12, Gig1/0/13,
Gig1/0/14
                                                Gig1/0/15, Gig1/0/16, Gig1/0/17,
Gig1/0/18
                                                Gig1/0/19, Gig1/0/20, Gig1/1/1, Gig1/1/2
                                                Gig1/1/3, Gig1/1/4
10   Manager_Vlan                     active
20   Staff_Vlan                       active
30   Security_Vlan                    active
40   Shop_Vlan                        active
50   VoIP                             active
60   Wireless                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
ML-S2#
```

VLAN TRUNKING

Configure vlan trunking in MLS1 and MLS2 which allows a single physical link between switches

```
ML-S1>en
ML-S1#sh vtp sta
ML-S1#sh vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 :
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0004.9A56.0500
Configuration last modified by 0.0.0.0 at 3-1-93 00:00:00
Local updater ID is 172.16.10.3 on interface Vl10 (lowest numbered VLAN interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 11
Configuration Revision            : 30
MD5 digest                        : 0xBA 0x9F 0x3E 0x90 0x11 0x8D 0x97 0xBE
                                    0x04 0x7D 0xAF 0x0D 0xDD 0xD6 0xE7 0x56
ML-S1#
```

```
ML-S2>en
ML-S2#sh vtp st
ML-S2#sh vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 :
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0001.64C3.41C0
Configuration last modified by 0.0.0.0 at 3-1-93 00:00:00
Local updater ID is 172.16.10.2 on interface Vl10 (lowest numbered VLAN interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 11
Configuration Revision            : 30
MD5 digest                        : 0xBA 0x9F 0x3E 0x90 0x11 0x8D 0x97 0xBE
                                    0x04 0x7D 0xAF 0x0D 0xDD 0xD6 0xE7 0x56
ML-S2#
```

OSPF

Configure OSPF in both multilayer switches and routers which we use to exchange routing information between devices in our network

```
ML-S1#sh ip rou ospf
     172.16.0.0/16 is variably subnetted, 16 subnets, 3 masks
O       172.16.2.0 [110/2] via 172.16.10.2, 06:05:46, Vlan10
                   [110/2] via 172.16.20.2, 06:05:46, Vlan20
                   [110/2] via 172.16.30.2, 06:05:46, Vlan30
                   [110/2] via 172.16.40.2, 06:05:46, Vlan40
                   [110/2] via 172.16.50.2, 06:05:46, Vlan50
                   [110/2] via 172.16.60.2, 06:05:46, Vlan60
                   [110/2] via 172.16.4.1, 06:05:46, GigabitEthernet1/0/2
O       172.16.3.0 [110/2] via 172.16.10.2, 06:05:46, Vlan10
                   [110/2] via 172.16.20.2, 06:05:46, Vlan20
                   [110/2] via 172.16.30.2, 06:05:46, Vlan30
                   [110/2] via 172.16.40.2, 06:05:46, Vlan40
                   [110/2] via 172.16.50.2, 06:05:46, Vlan50
                   [110/2] via 172.16.60.2, 06:05:46, Vlan60
                   [110/2] via 172.16.1.1, 06:05:46, GigabitEthernet1/0/1
O       172.16.6.0 [110/67] via 172.16.1.1, 06:05:46, GigabitEthernet1/0/1
                   [110/67] via 172.16.4.1, 06:05:46, GigabitEthernet1/0/2
O       172.16.7.0 [110/2] via 172.16.10.2, 06:05:46, Vlan10
                   [110/2] via 172.16.20.2, 06:05:46, Vlan20
                   [110/2] via 172.16.30.2, 06:05:46, Vlan30
                   [110/2] via 172.16.40.2, 06:05:46, Vlan40
                   [110/2] via 172.16.50.2, 06:05:46, Vlan50

ML-S1#
```

```
ML-S2#sh ip rou ospf
     172.16.0.0/16 is variably subnetted, 17 subnets, 3 masks
O       172.16.0.0 [110/2] via 172.16.10.3, 06:06:11, Vlan10
                   [110/2] via 172.16.20.3, 06:06:11, Vlan20
                   [110/2] via 172.16.30.3, 06:06:11, Vlan30
                   [110/2] via 172.16.40.3, 06:06:11, Vlan40
                   [110/2] via 172.16.50.3, 06:06:11, Vlan50
                   [110/2] via 172.16.60.3, 06:06:11, Vlan60
O       172.16.1.0 [110/2] via 172.16.10.3, 06:06:11, Vlan10
                   [110/2] via 172.16.20.3, 06:06:11, Vlan20
                   [110/2] via 172.16.30.3, 06:06:11, Vlan30
                   [110/2] via 172.16.40.3, 06:06:11, Vlan40
                   [110/2] via 172.16.50.3, 06:06:11, Vlan50
                   [110/2] via 172.16.60.3, 06:06:11, Vlan60
                   [110/2] via 172.16.3.1, 06:06:11, GigabitEthernet1/0/2
O       172.16.4.0 [110/2] via 172.16.10.3, 06:06:11, Vlan10
                   [110/2] via 172.16.20.3, 06:06:11, Vlan20
                   [110/2] via 172.16.30.3, 06:06:11, Vlan30
                   [110/2] via 172.16.40.3, 06:06:11, Vlan40
                   [110/2] via 172.16.50.3, 06:06:11, Vlan50
                   [110/2] via 172.16.60.3, 06:06:11, Vlan60
                   [110/2] via 172.16.2.1, 06:06:11, GigabitEthernet1/0/1
O       172.16.5.0 [110/2] via 172.16.10.3, 06:06:11, Vlan10

ML-S2#
ML-S2#
```

OSPF

```
Zone-1#sh ip rou os
Zone-1#sh ip rou ospf
     172.16.0.0/16 is variably subnetted, 18 subnets, 4 masks
O       172.16.0.0 [110/2] via 172.16.1.2, 10:08:11, FastEthernet0/1
O       172.16.2.0 [110/2] via 172.16.3.2, 10:08:11, FastEthernet1/0
O       172.16.4.0 [110/2] via 172.16.1.2, 06:07:09, FastEthernet0/1
O       172.16.5.0 [110/2] via 172.16.1.2, 10:08:11, FastEthernet0/1
O       172.16.6.0 [110/66] via 209.165.3.1, 06:07:09, FastEthernet0/0
                   [110/66] via 209.165.6.1, 06:07:09, FastEthernet1/1
O       172.16.7.0 [110/2] via 172.16.3.2, 10:08:11, FastEthernet1/0
O       172.16.8.0 [110/2] via 172.16.3.2, 10:08:11, FastEthernet1/0
O       172.16.9.0 [110/2] via 172.16.1.2, 10:08:11, FastEthernet0/1
O       172.16.10.0 [110/2] via 172.16.1.2, 10:06:45, FastEthernet0/1
                    [110/2] via 172.16.3.2, 10:06:45, FastEthernet1/0
O       172.16.20.0 [110/2] via 172.16.1.2, 06:07:09, FastEthernet0/1
                    [110/2] via 172.16.3.2, 06:07:09, FastEthernet1/0
O       172.16.30.0 [110/2] via 172.16.1.2, 10:06:09, FastEthernet0/1
                    [110/2] via 172.16.3.2, 10:06:09, FastEthernet1/0
O       172.16.40.0 [110/2] via 172.16.1.2, 10:07:32, FastEthernet0/1
                    [110/2] via 172.16.3.2, 10:07:32, FastEthernet1/0
O       172.16.50.0 [110/2] via 172.16.1.2, 10:06:09, FastEthernet0/1
                    [110/2] via 172.16.3.2, 10:06:09, FastEthernet1/0
O       172.16.60.0 [110/2] via 172.16.1.2, 10:08:11, FastEthernet0/1
                    [110/2] via 172.16.3.2, 10:08:11, FastEthernet1/0

Zone-1#
```

```
Zone-2>en
Zone-2#sh ip rou os
Zone-2#sh ip rou ospf
     172.16.0.0/16 is variably subnetted, 18 subnets, 4 masks
O       172.16.0.0 [110/2] via 172.16.4.2, 10:08:39, FastEthernet1/1
O       172.16.1.0 [110/2] via 172.16.4.2, 10:08:39, FastEthernet1/1
                   [110/2] via 172.16.2.2, 10:08:39, FastEthernet0/1
O       172.16.3.0 [110/2] via 172.16.2.2, 10:08:39, FastEthernet0/1
O       172.16.5.0 [110/2] via 172.16.4.2, 10:08:39, FastEthernet1/1
O       172.16.6.0 [110/66] via 209.165.5.1, 10:05:19, FastEthernet1/0
                   [110/66] via 209.165.4.1, 10:05:19, FastEthernet0/0
O       172.16.7.0 [110/2] via 172.16.2.2, 10:08:39, FastEthernet0/1
O       172.16.8.0 [110/2] via 172.16.2.2, 10:08:39, FastEthernet0/1
O       172.16.9.0 [110/2] via 172.16.4.2, 10:08:39, FastEthernet1/1
O       172.16.10.0 [110/2] via 172.16.2.2, 10:07:12, FastEthernet0/1
                    [110/2] via 172.16.4.2, 10:07:12, FastEthernet1/1
O       172.16.20.0 [110/2] via 172.16.2.2, 06:07:37, FastEthernet0/1
                    [110/2] via 172.16.4.2, 06:07:37, FastEthernet1/1
O       172.16.30.0 [110/2] via 172.16.2.2, 10:06:37, FastEthernet0/1
                    [110/2] via 172.16.4.2, 10:06:37, FastEthernet1/1
O       172.16.40.0 [110/2] via 172.16.2.2, 10:08:12, FastEthernet0/1
                    [110/2] via 172.16.4.2, 10:08:12, FastEthernet1/1
O       172.16.50.0 [110/2] via 172.16.2.2, 10:07:12, FastEthernet0/1
                    [110/2] via 172.16.4.2, 10:07:12, FastEthernet1/1
O       172.16.60.0 [110/2] via 172.16.2.2, 10:08:00, FastEthernet0/1

Zone-2#
Zone-2#
Zone-2#
```

STATIC ROUTES

Configure static routes on routers to send default gateway if the router does not have a specific route to the destination

```
Zone-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 209.165.3.1 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 18 subnets, 4 masks
O       172.16.0.0/26 [110/2] via 172.16.1.2, 10:22:28, FastEthernet0/1
C       172.16.1.0/30 is directly connected, FastEthernet0/1
L       172.16.1.1/32 is directly connected, FastEthernet0/1
O       172.16.2.0/30 [110/2] via 172.16.3.2, 10:22:28, FastEthernet1/0
C       172.16.3.0/30 is directly connected, FastEthernet1/0
L       172.16.3.1/32 is directly connected, FastEthernet1/0
O       172.16.4.0/30 [110/2] via 172.16.1.2, 06:21:26, FastEthernet0/1
O       172.16.5.0/26 [110/2] via 172.16.1.2, 10:22:28, FastEthernet0/1
O       172.16.6.0/30 [110/66] via 209.165.3.1, 06:21:26, FastEthernet0/0
                      [110/66] via 209.165.6.1, 06:21:26, FastEthernet1/1
O       172.16.7.0/26 [110/2] via 172.16.3.2, 10:22:28, FastEthernet1/0
O       172.16.8.0/26 [110/2] via 172.16.3.2, 10:22:28, FastEthernet1/0
O       172.16.9.0/26 [110/2] via 172.16.1.2, 10:22:28, FastEthernet0/1
O       172.16.10.0/24 [110/2] via 172.16.1.2, 10:21:02, FastEthernet0/1
                       [110/2] via 172.16.3.2, 10:21:02, FastEthernet1/0
O       172.16.20.0/24 [110/2] via 172.16.1.2, 06:21:26, FastEthernet0/1
                       [110/2] via 172.16.3.2, 06:21:26, FastEthernet1/0
O       172.16.30.0/24 [110/2] via 172.16.1.2, 10:20:26, FastEthernet0/1
                       [110/2] via 172.16.3.2, 10:20:26, FastEthernet1/0
O       172.16.40.0/24 [110/2] via 172.16.1.2, 10:21:49, FastEthernet0/1
                       [110/2] via 172.16.3.2, 10:21:49, FastEthernet1/0
O       172.16.50.0/24 [110/2] via 172.16.1.2, 10:20:26, FastEthernet0/1
```

```
O       172.16.10.0/24 [110/2] via 172.16.1.2, 10:21:02, FastEthernet0/1
                       [110/2] via 172.16.3.2, 10:21:02, FastEthernet1/0
O       172.16.20.0/24 [110/2] via 172.16.1.2, 06:21:26, FastEthernet0/1
                       [110/2] via 172.16.3.2, 06:21:26, FastEthernet1/0
O       172.16.30.0/24 [110/2] via 172.16.1.2, 10:20:26, FastEthernet0/1
                       [110/2] via 172.16.3.2, 10:20:26, FastEthernet1/0
O       172.16.40.0/24 [110/2] via 172.16.1.2, 10:21:49, FastEthernet0/1
                       [110/2] via 172.16.3.2, 10:21:49, FastEthernet1/0
O       172.16.50.0/24 [110/2] via 172.16.1.2, 10:20:26, FastEthernet0/1
                       [110/2] via 172.16.3.2, 10:20:26, FastEthernet1/0
O       172.16.60.0/24 [110/2] via 172.16.1.2, 10:22:28, FastEthernet0/1
                       [110/2] via 172.16.3.2, 10:22:28, FastEthernet1/0
     209.165.1.0/30 is subnetted, 1 subnets
O       209.165.1.0/30 [110/65] via 209.165.3.1, 06:21:26, FastEthernet0/0
     209.165.2.0/30 is subnetted, 1 subnets
O       209.165.2.0/30 [110/65] via 209.165.6.1, 10:22:28, FastEthernet1/1
     209.165.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.3.0/30 is directly connected, FastEthernet0/0
L       209.165.3.2/32 is directly connected, FastEthernet0/0
     209.165.4.0/30 is subnetted, 1 subnets
O       209.165.4.0/30 [110/2] via 209.165.6.1, 10:22:28, FastEthernet1/1
     209.165.5.0/30 is subnetted, 1 subnets
O       209.165.5.0/30 [110/2] via 209.165.3.1, 06:21:26, FastEthernet0/0
     209.165.6.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.6.0/30 is directly connected, FastEthernet1/1
L       209.165.6.2/32 is directly connected, FastEthernet1/1
S*   0.0.0.0/0 [1/0] via 209.165.3.1

Zone-1#
```

```
Zone-2#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 209.165.4.1 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 18 subnets, 4 masks
O       172.16.0.0/26 [110/2] via 172.16.4.2, 10:23:38, FastEthernet1/1
O       172.16.1.0/30 [110/2] via 172.16.4.2, 10:23:38, FastEthernet1/1
                      [110/2] via 172.16.2.2, 10:23:38, FastEthernet0/1
C       172.16.2.0/30 is directly connected, FastEthernet0/1
L       172.16.2.1/32 is directly connected, FastEthernet0/1
O       172.16.3.0/30 [110/2] via 172.16.2.2, 10:23:38, FastEthernet0/1
C       172.16.4.0/30 is directly connected, FastEthernet1/1
L       172.16.4.1/32 is directly connected, FastEthernet1/1
O       172.16.5.0/26 [110/2] via 172.16.4.2, 10:23:38, FastEthernet1/1
O       172.16.6.0/30 [110/66] via 209.165.5.1, 10:20:18, FastEthernet1/0
                      [110/66] via 209.165.4.1, 10:20:18, FastEthernet0/0
O       172.16.7.0/26 [110/2] via 172.16.2.2, 10:23:38, FastEthernet0/1
O       172.16.8.0/26 [110/2] via 172.16.2.2, 10:23:38, FastEthernet0/1
O       172.16.9.0/26 [110/2] via 172.16.4.2, 10:23:38, FastEthernet1/1
O       172.16.10.0/24 [110/2] via 172.16.2.2, 10:22:11, FastEthernet0/1
                       [110/2] via 172.16.4.2, 10:22:11, FastEthernet1/1
O       172.16.20.0/24 [110/2] via 172.16.2.2, 06:22:36, FastEthernet0/1
                       [110/2] via 172.16.4.2, 06:22:36, FastEthernet1/1
O       172.16.30.0/24 [110/2] via 172.16.2.2, 10:21:36, FastEthernet0/1
                       [110/2] via 172.16.4.2, 10:21:36, FastEthernet1/1
```

```
                       [110/2] via 172.16.4.2, 10:22:11, FastEthernet1/1
O       172.16.20.0/24 [110/2] via 172.16.2.2, 06:22:36, FastEthernet0/1
                       [110/2] via 172.16.4.2, 06:22:36, FastEthernet1/1
O       172.16.30.0/24 [110/2] via 172.16.2.2, 10:21:36, FastEthernet0/1
                       [110/2] via 172.16.4.2, 10:21:36, FastEthernet1/1
O       172.16.40.0/24 [110/2] via 172.16.2.2, 10:23:11, FastEthernet0/1
                       [110/2] via 172.16.4.2, 10:23:11, FastEthernet1/1
O       172.16.50.0/24 [110/2] via 172.16.2.2, 10:22:11, FastEthernet0/1
                       [110/2] via 172.16.4.2, 10:22:11, FastEthernet1/1
O       172.16.60.0/24 [110/2] via 172.16.2.2, 10:22:59, FastEthernet0/1
                       [110/2] via 172.16.4.2, 10:22:59, FastEthernet1/1
     209.165.1.0/30 is subnetted, 1 subnets
O       209.165.1.0/30 [110/65] via 209.165.5.1, 10:23:38, FastEthernet1/0
     209.165.2.0/30 is subnetted, 1 subnets
O       209.165.2.0/30 [110/65] via 209.165.4.1, 10:23:38, FastEthernet0/0
     209.165.3.0/30 is subnetted, 1 subnets
O       209.165.3.0/30 [110/2] via 209.165.5.1, 10:23:38, FastEthernet1/0
     209.165.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.4.0/30 is directly connected, FastEthernet0/0
L       209.165.4.2/32 is directly connected, FastEthernet0/0
     209.165.5.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.5.0/30 is directly connected, FastEthernet1/0
L       209.165.5.2/32 is directly connected, FastEthernet1/0
     209.165.6.0/30 is subnetted, 1 subnets
O       209.165.6.0/30 [110/2] via 209.165.4.1, 10:23:38, FastEthernet0/0
S*   0.0.0.0/0 [1/0] via 209.165.4.1

Zone-2#
```

STATIC ROUTES

PE Airport

```
PE-Airport>en
PE-Airport#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 209.165.2.2 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 17 subnets, 4 masks
O       172.16.0.0/26 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
                      [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
S       172.16.1.0/30 is directly connected, Serial0/0/0
S       172.16.2.0/30 is directly connected, Serial0/0/1
O       172.16.3.0/30 [110/66] via 209.165.2.2, 00:58:16, Serial0/0/1
                      [110/66] via 209.165.1.2, 00:58:16, Serial0/0/0
O       172.16.4.0/30 [110/66] via 209.165.2.2, 00:58:16, Serial0/0/1
                      [110/66] via 209.165.1.2, 00:58:16, Serial0/0/0
O       172.16.5.0/26 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
                      [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
C       172.16.6.0/30 is directly connected, FastEthernet0/0
L       172.16.6.1/32 is directly connected, FastEthernet0/0
O       172.16.7.0/26 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
                      [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
O       172.16.8.0/26 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
                      [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
O       172.16.9.0/26 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
                      [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
O       172.16.10.0/24 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
                       [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
O       172.16.20.0/24 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
```

```
                       [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
O       172.16.30.0/24 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
                       [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
O       172.16.40.0/24 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
                       [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
O       172.16.50.0/24 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
                       [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
O       172.16.60.0/24 [110/67] via 209.165.2.2, 00:58:16, Serial0/0/1
                       [110/67] via 209.165.1.2, 00:58:16, Serial0/0/0
     209.165.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.1.0/30 is directly connected, Serial0/0/0
L       209.165.1.1/32 is directly connected, Serial0/0/0
     209.165.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.2.0/30 is directly connected, Serial0/0/1
L       209.165.2.1/32 is directly connected, Serial0/0/1
     209.165.3.0/30 is subnetted, 1 subnets
O       209.165.3.0/30 [110/65] via 209.165.1.2, 00:58:16, Serial0/0/0
     209.165.4.0/30 is subnetted, 1 subnets
O       209.165.4.0/30 [110/65] via 209.165.2.2, 06:23:17, Serial0/0/1
     209.165.5.0/30 is subnetted, 1 subnets
O       209.165.5.0/30 [110/65] via 209.165.1.2, 00:58:16, Serial0/0/0
     209.165.6.0/30 is subnetted, 1 subnets
O       209.165.6.0/30 [110/65] via 209.165.2.2, 06:23:17, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 209.165.2.2, 00:58:16, Serial0/0/1
               [110/1] via 209.165.1.2, 00:58:16, Serial0/0/0

PE-Airport#
```

HSRP

Configure hsrp on multilayer switches to provide redundancy on our network.

```
ML-S1#sh standby br
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   100   Active   local           172.16.10.2     172.16.10.1
Vl20        20   100   Active   local           172.16.20.2     172.16.20.1
Vl30        30   100   Active   local           172.16.30.2     172.16.30.1
Vl40        40   100   Active   local           172.16.40.2     172.16.40.1
Vl50        50   100   Active   local           172.16.50.2     172.16.50.1
Vl60        60   100   Active   local           172.16.60.2     172.16.60.1
ML-S1#
```

```
ML-S2# sh stand
ML-S2# sh standby
ML-S2# sh standby br
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   100   Standby  172.16.10.3     local           172.16.10.1
Vl20        20   100   Standby  172.16.20.3     local           172.16.20.1
Vl30        30   100   Standby  172.16.30.3     local           172.16.30.1
Vl40        40   100   Standby  172.16.40.3     local           172.16.40.1
Vl50        50   100   Standby  172.16.50.3     local           172.16.50.1
Vl60        60   100   Standby  172.16.60.3     local           172.16.60.1
ML-S2#
```

ETHERCHANNEL

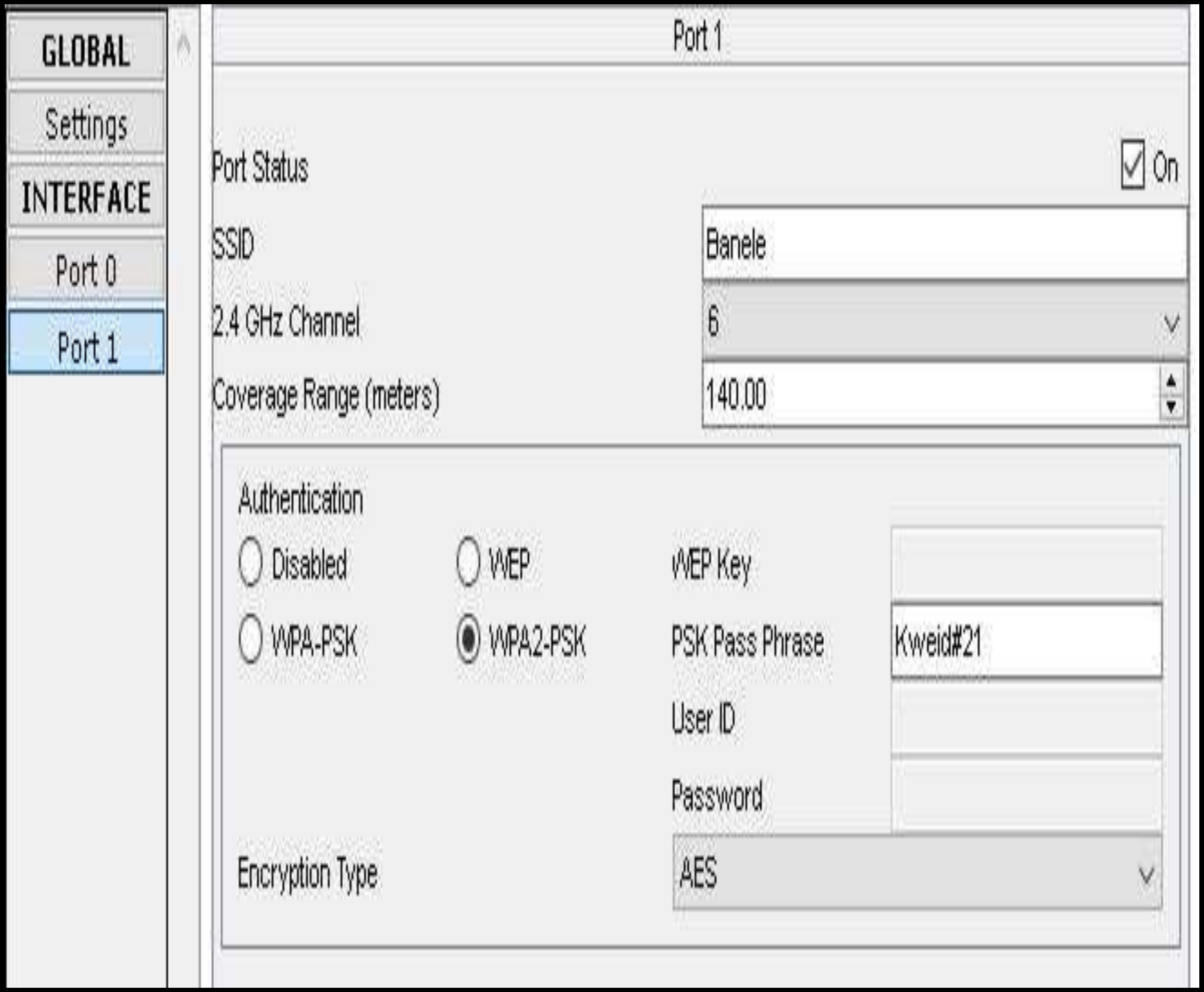
Configure etherchannel on my multilayer switches to combine the multiple links between our multilayer switches into single local link which increase the redundancy bandwidth.

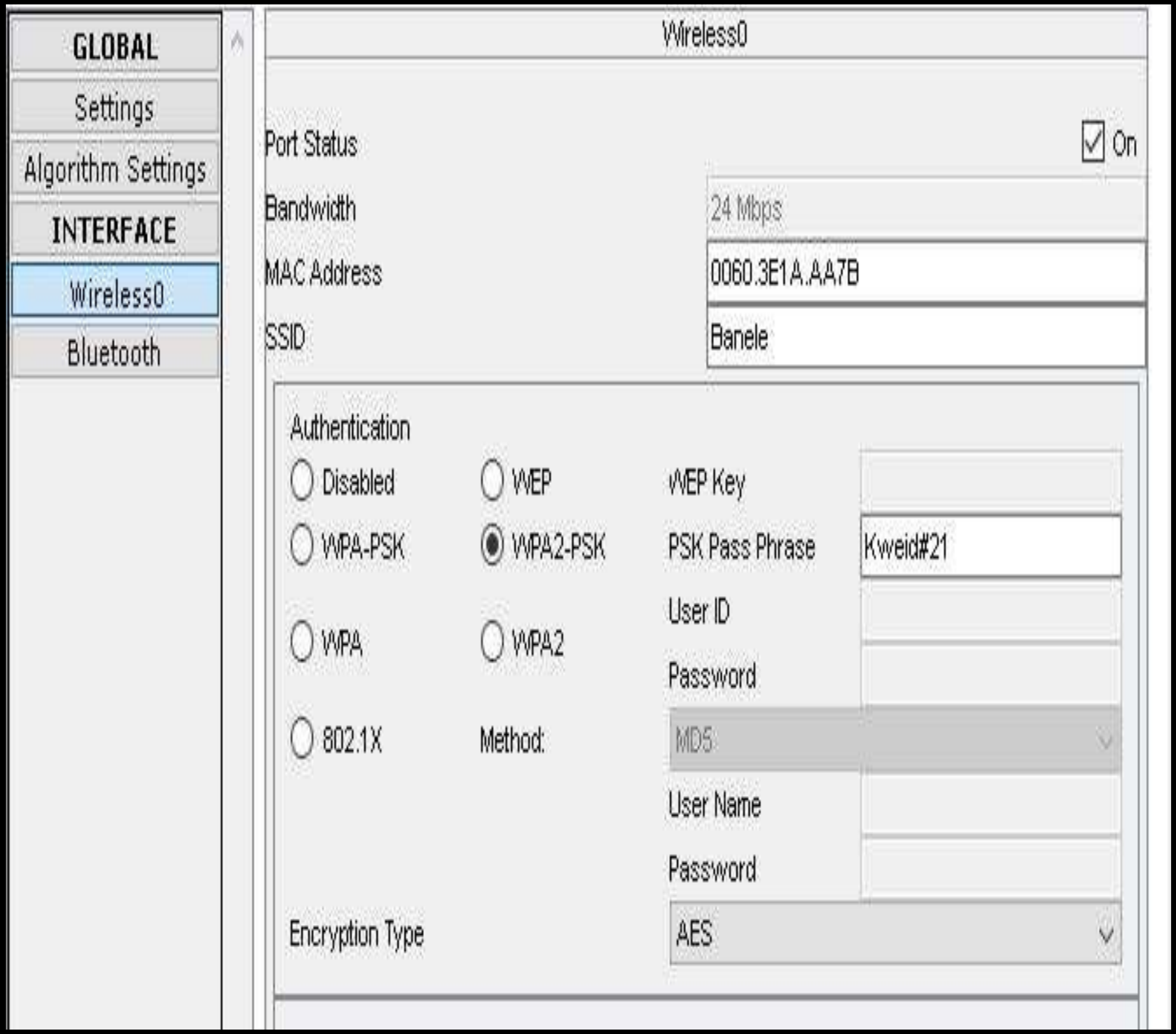
```
V150         50   100  Active   local            172.16.50.2     172.16.50.1
V160         60   100  Active   local            172.16.60.2     172.16.60.1
ML-S1#sh et
ML-S1#sh etherchannel s
ML-S1#sh etherchannel summary
ML-S1#sh etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Gig1/0/21(P) Gig1/0/22(P) Gig1/0/23(P) Gig1/0/24(P)
ML-S1#
```

```
V150         50   100  Standby 172.16.50.3     local           172.16.50.1
V160         60   100  Standby 172.16.60.3     local           172.16.60.1
ML-S2#sh et
ML-S2#sh etherchannel s
ML-S2#sh etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Gig1/0/21(P) Gig1/0/22(P) Gig1/0/23(P) Gig1/0/24(P)
ML-S2#
ML-S2#
```

WIRELESS CONFIG

We will access point for wireless config to allow wireless devices to connect to a wired network.





PAT

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 209.165.1.1

Pinging 209.165.1.1 with 32 bytes of data:

Reply from 209.165.1.1: bytes=32 time=14ms TTL=252
Reply from 209.165.1.1: bytes=32 time=12ms TTL=252
Reply from 209.165.1.1: bytes=32 time=11ms TTL=252
Reply from 209.165.1.1: bytes=32 time=12ms TTL=252

Ping statistics for 209.165.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 14ms, Average = 12ms

C:\>ping 209.165.2.1

Pinging 209.165.2.1 with 32 bytes of data:

Reply from 209.165.2.1: bytes=32 time=23ms TTL=252
Reply from 209.165.2.1: bytes=32 time=15ms TTL=252
Reply from 209.165.2.1: bytes=32 time=104ms TTL=252
Reply from 209.165.2.1: bytes=32 time=1ms TTL=252

Ping statistics for 209.165.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 104ms, Average = 35ms

C:\>
```

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 209.165.1.1

Pinging 209.165.1.1 with 32 bytes of data:

Reply from 209.165.1.1: bytes=32 time=41ms TTL=252
Reply from 209.165.1.1: bytes=32 time=53ms TTL=252
Reply from 209.165.1.1: bytes=32 time=53ms TTL=252
Reply from 209.165.1.1: bytes=32 time=59ms TTL=252

Ping statistics for 209.165.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 41ms, Maximum = 59ms, Average = 51ms

C:\>ping 209.165.2.1

Pinging 209.165.2.1 with 32 bytes of data:

Reply from 209.165.2.1: bytes=32 time=34ms TTL=252
Reply from 209.165.2.1: bytes=32 time=174ms TTL=252
Reply from 209.165.2.1: bytes=32 time=16ms TTL=252
Reply from 209.165.2.1: bytes=32 time=62ms TTL=252

Ping statistics for 209.165.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 174ms, Average = 71ms

C:\>
```

PAT

We configured PAT on both border routers to allow multiple devices on private network to share a single public IP address and port number.

```
Zone-1#sh ip nat translations
Pro  Inside global      Inside local     Outside local    Outside global
icmp 209.165.3.2:1024   172.16.60.5:1    209.165.2.1:1    209.165.2.1:1024
icmp 209.165.3.2:1      172.16.40.5:1    209.165.1.1:1    209.165.1.1:1
icmp 209.165.3.2:2      172.16.20.4:2    209.165.2.1:2    209.165.2.1:2
icmp 209.165.3.2:3      172.16.20.4:3    209.165.1.1:3    209.165.1.1:3

Zone-1#
```

```
Zone-2#sh ip nat tr
Zone-2#sh ip nat translations
Pro  Inside global      Inside local     Outside local    Outside global
icmp 209.165.4.2:1024   172.16.30.4:1    209.165.1.1:1    209.165.1.1:1024
icmp 209.165.4.2:18     172.16.4.2:18    209.165.1.1:18   209.165.1.1:18
icmp 209.165.4.2:1      172.16.20.4:1    209.165.1.1:1    209.165.1.1:1
icmp 209.165.4.2:20     172.16.4.2:20    209.165.1.1:20   209.165.1.1:20
icmp 209.165.4.2:54     172.16.2.2:54    209.165.1.1:54   209.165.1.1:54
icmp 209.165.4.2:56     172.16.2.2:56    209.165.1.1:56   209.165.1.1:56

Zone-2#
Zone-2#
```

VPN

We configured VPN on both our border routers and PE Airport router, what will allow secure access border routers from a PE Airport router

```
Zone-1#sh crypto isa
Zone-1#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
209.165.1.1     209.165.3.2     QM_IDLE           1096    0 ACTIVE


IPv6 Crypto ISAKMP SA


Zone-1#
```

```
Zone-2#SH CR
Zone-2#SH CRypto ISA
Zone-2#SH CRypto ISAkmp SA
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
209.165.2.1     209.165.4.2     QM_IDLE           1044    0 ACTIVE


IPv6 Crypto ISAKMP SA


Zone-2#
```

VPN

```
PE-Airport#ping
Protocol [ip]:
PE-Airport#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
209.165.3.2     209.165.1.1     QM_IDLE           1042    0 ACTIVE

209.165.4.2     209.165.2.1     QM_IDLE           1039    0 ACTIVE


IPv6 Crypto ISAKMP SA


PE-Airport#
```

ZBF

We configured ZBF on our border routers which will block devices unless explicitly allowed.

```
Zone-1#sh zone-
Zone-1#sh zone-pair se
Zone-1#sh zone-pair security ?
  <cr>
Zone-1#sh zone-pair security
Zone-pair name INSIDE-1_TO_OUTSIDE-1
    Source-Zone INSIDE-1  Destination-Zone OUTSIDE-1
    service-policy INSIDE-1_TO_OUTSIDE-1

Zone-pair name INSIDE-2_TO_OUTSIDE-2
    Source-Zone INSIDE-2  Destination-Zone OUTSIDE-2
    service-policy INSIDE-2_TO_OUTSIDE-2

Zone-pair name INSIDE-1_TO_OUTSIDE-2
    Source-Zone INSIDE-1  Destination-Zone OUTSIDE-2
    service-policy INSIDE-1_TO_OUTSIDE-2

Zone-pair name INSIDE-2_TO_OUTSIDE-1
    Source-Zone INSIDE-2  Destination-Zone OUTSIDE-1
    service-policy INSIDE-2_TO_OUTSIDE-1
```

```
  security   Security Zone
Zone-2#sh zone-pair se
Zone-2#sh zone-pair security ?
  <cr>
Zone-2#sh zone-pair security
Zone-pair name INSIDE-1_TO_OUTSIDE-1
    Source-Zone INSIDE-1  Destination-Zone OUTSIDE-1
    service-policy INSIDE-1_TO_OUTSIDE-1

Zone-pair name INSIDE-1_TO_OUTSIDE-2
    Source-Zone INSIDE-1  Destination-Zone OUTSIDE-2
    service-policy INSIDE-1_TO_OUTSIDE-2

Zone-pair name INSIDE-2_TO_OUTSIDE-2
    Source-Zone INSIDE-2  Destination-Zone OUTSIDE-2
    service-policy INSIDE-2_TO_OUTSIDE-2

Zone-pair name INSIDE-2_TO_OUTSIDE-1
    Source-Zone INSIDE-2  Destination-Zone OUTSIDE-1
    service-policy INSIDE-2_TO_OUTSIDE-1
```

ZBF

For Zone-1

```
      0 packets, 0 bytes
Zone-1#sh policy-map type inspect zone-pair sessions

policy exists on zp INSIDE-1_TO_OUTSIDE-1
 Zone-pair: INSIDE-1_TO_OUTSIDE-1

  Service-policy inspect : INSIDE-1_TO_OUTSIDE-1

    Class-map: INSIDE-1_PROTOCOLS (match-any)
      Match: protocol icmp
        3 packets, 384 bytes
        30 second rate 0 bps
      Match: protocol tcp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol udp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: access-group 150
        0 packets, 0 bytes
        30 second rate 0 bps
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes

policy exists on zp INSIDE-2_TO_OUTSIDE-2
 Zone-pair: INSIDE-2_TO_OUTSIDE-2

  Service-policy inspect : INSIDE-2_TO_OUTSIDE-2

    Class-map: INSIDE-2_PROTOCOLS (match-any)
      Match: protocol dns
        0 packets, 0 bytes
```

```
    Class-map: INSIDE-2_PROTOCOLS (match-any)
      Match: protocol dns
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol http
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol https
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: access-group 150
        0 packets, 0 bytes
        30 second rate 0 bps
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes

policy exists on zp INSIDE-1_TO_OUTSIDE-2
 Zone-pair: INSIDE-1_TO_OUTSIDE-2

  Service-policy inspect : INSIDE-1_TO_OUTSIDE-2

    Class-map: INSIDE-1_PROTOCOLS (match-any)
      Match: protocol icmp
        3 packets, 384 bytes
        30 second rate 0 bps
      Match: protocol tcp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol udp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: access-group 150
```

```
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: access-group 150
        0 packets, 0 bytes
        30 second rate 0 bps
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes

policy exists on zp INSIDE-2_TO_OUTSIDE-1
 Zone-pair: INSIDE-2_TO_OUTSIDE-1

  Service-policy inspect : INSIDE-2_TO_OUTSIDE-1

    Class-map: INSIDE-2_PROTOCOLS (match-any)
      Match: protocol dns
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol http
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol https
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: access-group 150
        0 packets, 0 bytes
        30 second rate 0 bps
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
Zone-1#
```

ZBF

For Zone-2

```
Zone-2#sh policy-map type inspect zone-pair S
Zone-2#sh policy-map type inspect zone-pair Sessions

policy exists on zp INSIDE-1_TO_OUTSIDE-1
 Zone-pair: INSIDE-1_TO_OUTSIDE-1

  Service-policy inspect : INSIDE-1_TO_OUTSIDE-1

    Class-map: INSIDE-1_PROTOCOLS (match-any)
      Match: protocol icmp
        9 packets, 452 bytes
        30 second rate 0 bps
      Match: protocol tcp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol udp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: access-group 150
        0 packets, 0 bytes
        30 second rate 0 bps
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes

policy exists on zp INSIDE-1_TO_OUTSIDE-2
 Zone-pair: INSIDE-1_TO_OUTSIDE-2

  Service-policy inspect : INSIDE-1_TO_OUTSIDE-2

    Class-map: INSIDE-1_PROTOCOLS (match-any)
      Match: protocol icmp
        9 packets, 452 bytes
        30 second rate 0 bps
```

```
  Service-policy inspect : INSIDE-1_TO_OUTSIDE-2

    Class-map: INSIDE-1_PROTOCOLS (match-any)
      Match: protocol icmp
        9 packets, 452 bytes
        30 second rate 0 bps
      Match: protocol tcp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol udp
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: access-group 150
        0 packets, 0 bytes
        30 second rate 0 bps
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes

policy exists on zp INSIDE-2_TO_OUTSIDE-2
 Zone-pair: INSIDE-2_TO_OUTSIDE-2

  Service-policy inspect : INSIDE-2_TO_OUTSIDE-2

    Class-map: INSIDE-2_PROTOCOLS (match-any)
      Match: protocol dns
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol http
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol https
        0 packets, 0 bytes
        30 second rate 0 bps
```

```
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: access-group 150
        0 packets, 0 bytes
        30 second rate 0 bps
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes

policy exists on zp INSIDE-2_TO_OUTSIDE-1
 Zone-pair: INSIDE-2_TO_OUTSIDE-1

  Service-policy inspect : INSIDE-2_TO_OUTSIDE-1

    Class-map: INSIDE-2_PROTOCOLS (match-any)
      Match: protocol dns
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol http
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: protocol https
        0 packets, 0 bytes
        30 second rate 0 bps
      Match: access-group 150
        0 packets, 0 bytes
        30 second rate 0 bps
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
Zone-2#
```

VERYING PINGING BETWEEN VLANS

```
C:\>ping 172.16.40.5

Pinging 172.16.40.5 with 32 bytes of data:

Reply from 172.16.40.5: bytes=32 time=81ms TTL=127
Reply from 172.16.40.5: bytes=32 time=10ms TTL=127
Reply from 172.16.40.5: bytes=32 time=30ms TTL=127
Reply from 172.16.40.5: bytes=32 time=13ms TTL=127

Ping statistics for 172.16.40.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 81ms, Average = 33ms

C:\>ping 172.16.30.5

Pinging 172.16.30.5 with 32 bytes of data:

Reply from 172.16.30.5: bytes=32 time=25ms TTL=127
Reply from 172.16.30.5: bytes=32 time=15ms TTL=127
Reply from 172.16.30.5: bytes=32 time=12ms TTL=127
Reply from 172.16.30.5: bytes=32 time=16ms TTL=127

Ping statistics for 172.16.30.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 25ms, Average = 17ms

C:\>
```

PC9

Physical Config Desktop Programming Attributes

Command Prompt

```
Reply from 172.16.30.5: bytes=32 time=40ms TTL=127
Reply from 172.16.30.5: bytes=32 time=51ms TTL=127
Reply from 172.16.30.5: bytes=32 time=22ms TTL=127
Reply from 172.16.30.5: bytes=32 time=12ms TTL=127

Ping statistics for 172.16.30.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 51ms, Average = 31ms

C:\>ping 172.16.20.4

Pinging 172.16.20.4 with 32 bytes of data:

Reply from 172.16.20.4: bytes=32 time=12ms TTL=127
Reply from 172.16.20.4: bytes=32 time=15ms TTL=127
Reply from 172.16.20.4: bytes=32 time=12ms TTL=127
Reply from 172.16.20.4: bytes=32 time=15ms TTL=127

Ping statistics for 172.16.20.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 15ms, Average = 13ms

C:\>ping 172.16.30.5

Pinging 172.16.30.5 with 32 bytes of data:

Reply from 172.16.30.5: bytes=32 time=11ms TTL=127
Reply from 172.16.30.5: bytes=32 time=11ms TTL=127
Reply from 172.16.30.5: bytes=32 time=15ms TTL=127
Reply from 172.16.30.5: bytes=32 time=11ms TTL=127

Ping statistics for 172.16.30.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 15ms, Average = 12ms
```